CFA® 2026 Level I

Complete Study Guide

Quantitative Methods, Economics, and Corporate Issuers

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This document is a condensed study guide. Please refer to the official CFA Institute curriculum for complete details.

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Part I

Quantitative Methods

1 Rates and Returns

1.1 Interest Rates and Return Measurement

LOS 1.a: Interpret interest rates as required rates of return, discount rates, or opportunity costs and explain an interest rate as the sum of a real risk-free rate and premiums that compensate investors for bearing distinct types of risk.

Why This Matters for the CFA Exam: Interest rates are the foundation of all valuation. Every model you learn; bond valuation, stock valuation, project NPV; uses interest rates to discount cash flows. Understanding the components of interest rates helps you predict how rates change when economic conditions change.

Three Interpretations of an Interest Rate:

1. **Required Rate of Return:** The minimum return an investor demands to invest. If a bond offers 5% but you require 6%, you will not buy it unless the price drops.

2. **Discount Rate:** Used to convert future cash flows into present value. A higher discount rate means lower present value.

3. **Opportunity Cost:** The benefit forgone by consuming today instead of saving. If you spend $100 today instead of investing it at 5%, you lose $5 of future consumption.

The Build-Up Method (Components of an Interest Rate):

$$\boxed{\begin{gathered}\text{Nominal Rate} = \text{Real Risk-Free Rate} + \text{Inflation Premium} + \\ \text{Default Risk Premium} + \text{Liquidity Premium} + \\ \text{Maturity Premium}\end{gathered}}$$

Component	Description
Real Risk-Free Rate	Compensation for postponing consumption (time preference). Assumes no inflation and no default risk. Estimated using TIPS yields.
Inflation Premium	Compensation for expected loss of purchasing power. Higher when inflation expectations rise.
Default Risk Premium	Compensation for risk that borrower fails to make promised payments. Higher for lower credit ratings (e.g., BBB vs AAA).
Liquidity Premium	Compensation for risk of selling at less than fair value quickly. Higher for assets with thin trading (e.g., small-cap stocks, real estate).
Maturity Premium	Compensation for greater price volatility of longer-term bonds. Longer maturity = higher interest rate risk.

Fisher Effect:

$$\boxed{\begin{gathered}(1 + \text{Nominal Risk-Free}) = (1 + \text{Real Risk-Free}) \\ \times (1 + \text{Expected Inflation})\end{gathered}}$$

Approximation (used in most CFA problems):

$$\text{Nominal} \approx \text{Real} + \text{Expected Inflation}$$

Example: If the real risk-free rate is 2% and expected inflation is 3%, the nominal risk-free rate is approximately 5% (exactly: 1.02 × 1.03 = 1.0506 → 5.06%).

Professor's Note: *Use the exact formula when inflation is high (e.g., >5%). Use the approximation for quick calculations on the exam.*

Exam Trick: On the exam, if they give you nominal and real rates, solve for implied inflation: Inflation $= \frac{1+\textbf{Nominal}}{1+\textbf{Real}} - 1$

1.2 Return Measurement Approaches

LOS 1.b: Calculate and interpret different approaches to return measurement over time and describe their appropriate uses.

Holding Period Return (HPR): The total return over a single period, including both price appreciation and cash flows.

$$\boxed{\text{HPR} = \text{P}_t + CF_t \overline{P_0 - 1 = \frac{\text{Ending Value} - \text{Beginning Value} + \text{Cash Flows}}{\text{Beginning Value}}}}$$

Example: Stock purchased for $50 pays $2 dividend and sells for $55:

$$\text{HPR} = \frac{55+2}{50} - 1 = \frac{57}{50} - 1 = 0.14 = 14\%$$

Multi-Period HPR: Chain returns geometrically:

$$\boxed{\text{HPR}_{1..n} = (1 + \text{HPR}_1) \times (1 + \text{HPR}_2) \times \ldots \times (1 + \text{HPR}_n) - 1}$$

Example: Returns: Year 1 = 10%, Year 2 = -5%, Year 3 = 8%:

$$\text{HPR} = (1.10)(0.95)(1.08) - 1 = 1.1286 - 1 = 12.86\%$$

Arithmetic Mean Return:

$$\boxed{= \text{R}_1 + R_2 + \ldots + R_n \overline{n}}$$

- **Use:** Forecasting next period's return (unbiased estimator).
- **Limitation:** Ignores compounding. Overstates true growth when returns vary.
- **Example:** Returns 100% and -50%. Arithmetic mean = 25%, but you actually broke even.

Geometric Mean Return (Compound Annual Rate):

$$\boxed{_G = \sqrt[n]{(1+R_1)(1+R_2)\ldots(1+R_n)} - 1}$$

- **Use:** Historical compound growth (e.g., "the fund returned 8% annually over 10 years").

- **Key Fact:** Geometric mean Arithmetic mean. Equality only when all returns identical.
- The difference increases with volatility.

Example: Returns: -9.34%, 23.45%, 8.92%:

$$\bar{R}_G = \sqrt[3]{(0.9066)(1.2345)(1.0892)} - 1 = \sqrt[3]{1.21903} - 1 = 6.825\%$$

Harmonic Mean:

$$\boxed{H = \frac{n}{\sum_{i=1}^{n} \frac{1}{X_i}}}$$

- **Use:** Average cost with dollar-cost averaging (fixed dollar amount each period).
- **Example:** Buy \$1,000 monthly at \$8, \$9, \$10:

$$\text{Harmonic mean} = \frac{3}{\frac{1}{8} + \frac{1}{9} + \frac{1}{10}} = \$8.93$$

- Total shares = 336.11. Average price = \$3,000/336.11 = \$8.93.

Harmonic <Geometric <Arithmetic (for non-constant values)

Summary Table:

Measure	Best Used For
Arithmetic Mean	Forecasting future returns, expected returns
Geometric Mean	Historical compound growth, performance comparison
Harmonic Mean	Dollar-cost averaging, average purchase price
Trimmed Mean	Reducing outlier effects (drop highest/lowest 5%)
Winsorized Mean	Reduce outliers without discarding (replace with next values)

1.3 Time-Weighted vs. Money-Weighted Returns

LOS 1.c: Compare the money-weighted and time-weighted rates of return and evaluate the performance of portfolios based on these measures.

Money-Weighted Return (MWR): The Internal Rate of Return of all cash flows into and out of the portfolio.

Calculation Steps:

1. Identify timing of each cash flow.
2. Deposits = inflows (positive). Withdrawals/ending value = outflows (negative).
3. Solve for IRR using calculator.

Example: MWR Calculation:

- t=0: Buy 1 share for \$100 (+100)
- t=1: Buy another for \$120 (+120), receive \$2 dividend (-2) → Net +118
- t=2: Sell both for \$260 (-260), receive \$4 dividend (-4) → Net -264

Calculator: CF0=+100, CF1=+118, CF2=-264, CPT IRR = 13.86%

Time-Weighted Return (TWR): Removes the effect of cash flow timing.

Calculation Steps:

1. Value portfolio before each cash flow.
2. Compute HPR for each subperiod.
3. Chain returns: $\prod(1 + \text{HPR}) - 1$
4. Annualize with geometric mean if ¿1 year.

Example: Same investment:

- Period 1: Begin \$100, end \$120, dividend \$2 → HPR = 22%

- Period 2: Begin \$240, end \$260, dividend \$4 → HPR = 10%
- TWR = $\sqrt{(1.22)(1.10)} - 1 = 15.84\%$

Comparison:

Feature	**MWR**	**TWR**
Affected by cash flow timing?	Yes	No
Preferred for manager evaluation?	No	Yes
Preferred when manager controls flows?	Yes	No
What does it measure?	Investor's return	Strategy return

Professor's Note: *If funds added before poor performance: MWR < TWR. If added before good performance: MWR > TWR.*

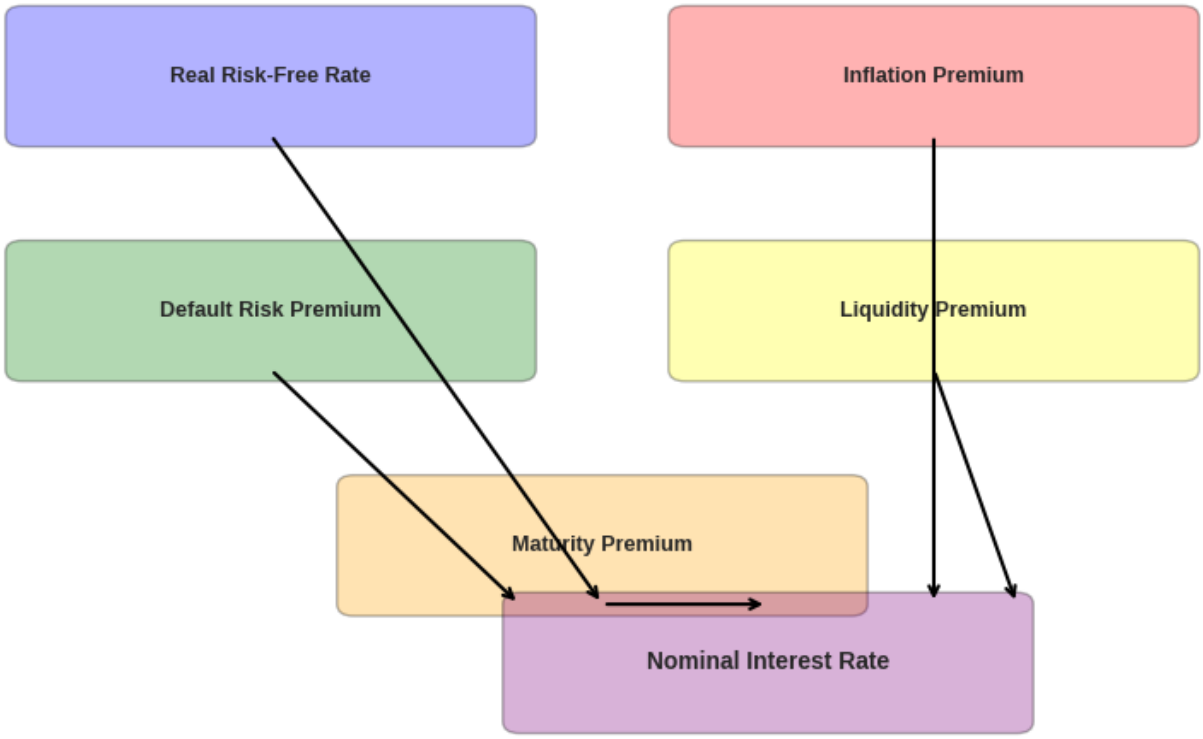

Figure 1: Components of a Nominal Interest Rate

1.4 Annualized and Continuously Compounded Returns

LOS 1.d: Calculate and interpret annualized return measures and continuously compounded returns.

Annualized Return:

$$\boxed{\text{Annualized Return} = (1 + \text{HPR})^{365/\text{days}} - 1}$$

Example: 90-day HPR of 0.75%:

$$\text{Annualized} = (1.0075)^{365/90} - 1 = 1.0075^{4.0556} - 1 = 3.08\%$$

Example: 500-day HPR of 3.09%:

$$\text{Annualized} = (1.0309)^{365/500} - 1 = 1.0309^{0.73} - 1 = 2.25\%$$

Continuously Compounded Return:

$$\boxed{\text{R}_{cc} = \ln(1 + \text{HPR}) = \ln\left(\frac{P_t}{P_0}\right)}$$

Example: Stock from \$20 to \$23:

$$R_{cc} = \ln(23/20) = \ln(1.15) = 13.98\%$$

Professor's Note: *Continuously compounded returns are additive across time. Used in Black-Scholes and derivatives pricing.*

Compounding Frequency:

$$FV = PV\left(1 + \frac{r}{m}\right)^{m \times n}$$

As m increases, FV increases. Continuous compounding gives maximum FV.

1.5 Major Return Measures

LOS 1.e: Calculate and interpret major return measures.

Measure	Definition
Gross Return	Before management/administration fees. Trading costs already deducted.
Net Return	After management/administration fees. What investor receives.
Pretax Nominal Return	Before taxes and inflation. The reported return.
After-Tax Nominal Return	After taxes, before inflation.
Real Return	Adjusted for inflation. Measures purchasing power increase.
Leveraged Return	Return on equity when borrowing amplifies results.

Real Return:

$$\boxed{\text{Real Return} = 1 + \text{Nominal}_{\frac{}{1+\text{Inflation}-1}}}$$

Example: Nominal = 7%, Inflation = 2%: Real = 1.07/1.02 - 1 = 4.9%

Leveraged Return:

$$\boxed{\text{Leveraged Return} = \text{r}(\text{V}_0 + V_B) - r_B V_{B\,\overline{V_0}}}$$

Example: Equity \$100k, borrow \$400k at 5%, earn 10% on \$500k:

$$\text{Return} = \frac{0.10(500k) - 0.05(400k)}{100k} = 30\%$$

Leverage amplifies both gains AND losses.

2 The Time Value of Money

2.1 Discounted Cash Flow Valuation

LOS 2.a: Calculate and interpret the present value of fixed-income and equity instruments based on expected future cash flows.

Basic TVM Relationship:

$$\boxed{\mathrm{FV} = \mathrm{PV}(1+\mathrm{r})^t \quad \text{or} \quad PV = \frac{FV}{(1+r)^t}}$$

Zero-Coupon Bond:

$$\boxed{\mathrm{PV} = \mathrm{FV}\frac{}{(1+r)^n}}$$

Example: Face value \$1,000, 15 years, 4% yield:

$$PV = \frac{1000}{(1.04)^{15}} = \$555.26$$

Fixed-Coupon Bond:

$$\boxed{\mathrm{PV} = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{FV}{(1+r)^n}}$$

Example: 10-year, \$1,000 par, 10% coupon, 8% yield: Calculator: N=10, PMT=100, FV=1000, I/Y=8, CPT PV = \$1,134.20

Perpetuity:

$$\boxed{\mathrm{PV} = \mathrm{PMT}\frac{}{r}}$$

Example: \$9 annual dividend, 11% required return: PV = 9/0.11 = \$81.82

Preferred Stock:

$$\boxed{\mathrm{V}_0 = \frac{D_p}{k_p}}$$

Constant Growth DDM (Gordon Growth):

$$\boxed{\mathrm{V}_0 = \frac{D_1}{k_e - g_c}}$$

(requires $\mathrm{k}_e > g_c$)

Example: D1 = \$1.62, g = 8%, $k_e = 12\% : V_0 = 1.62/(0.12-0.08) = \$40.50$

Multistage DDM:

1. *Calculate dividends during high-growth period.*
2. *Calculate terminal value using constant growth.*
3. *Discount all cash flows to present.*

Example: D0 = \$1.00, g = 15% for 2 years, then g = 5%,* $k_e = 11\%$:

D1 = 1.15, D2 = 1.32, D3 = 1.386

P2 = 1.386/(0.11-0.05) = 23.10

V0 = 1.15/1.11 + (1.32+23.10)/1.11^2 = \$20.86

2.2 Implied Returns and Cash Flow Additivity

LOS 2.b: Calculate and interpret implied return of fixed-income instruments and required return/implied growth of equity instruments.

Implied Return on Zero-Coupon Bond:

$$\mathrm{r} = \left(\frac{FV}{PV}\right)^{1/n} - 1$$

Example: PV = \$650, FV = \$1,000, n = 15: r = (1000/650)$^{1/15} - 1 =$ 2.91%

Implied Required Return on Equity (Constant Growth):

$$k_e = \frac{D_1}{V_0} + g_c$$

Implied Growth Rate:

$$g_c = k_e - \frac{D_1}{V_0}$$

Example: Price = \$40.50, D1 = \$1.62, $k_e = 12\% : g = 0.12 - 1.62/40.50 =$ 8%

LOS 2.c: Explain the cash flow additivity principle and its importance for the no-arbitrage condition.

Cash Flow Additivity Principle: *PV of total cash flows = Sum of PV of individual pieces.*

Example: Uneven cash flows: \$100, \$100, \$400, \$100 at 10%: Can replicate as 4-year annuity of \$100 + \$300 at t=3.

Forward Interest Rates:

$$(1 + S_2)^2 = (1 + S_1)(1 + 1y1y)$$

Example: S1 = 4%, S2 = 8%: 1y1y = $(1.08^2/1.04) - 1 = 12.154\%$

Forward Exchange Rates (No-Arbitrage):

$$F = S \times \frac{1+i_{price}}{1+i_{base}}$$

Example: S = 4.5671 ABE/DUB, $i_ABE = 5\%, i_DUB = 3\% : F = 4.5671 \times \frac{1.05}{1.03} = 4.6558$

3 Statistical Measures of Asset Returns

3.1 Central Tendency and Location

LOS 3.a: Calculate, interpret, and evaluate measures of central tendency and location.

Arithmetic Mean:

$$= \sum x_{i\frac{-}{n}}$$

Median: Middle value when data sorted. Resistant to outliers. *Example: Odd n: 30%, 25%, 23%, 21%, 15% → Median = 23% Example: Even n: Add 28% → 30%, 28%, 25%, 23%, 21%, 15% → Median = (25+23)/2 = 24%*

Mode: Most frequent value. Unimodal, bimodal, or trimodal.

Trimmed Mean: Excludes a stated percentage of extreme observations (e.g., 1% trimmed mean drops lowest 0.5% and highest 0.5%).

Winsorized Mean: Replaces extreme observations with next highest/lowest values.

Quantiles:

- Percentiles: 100 groups
- Deciles: 10 groups
- Quartiles: 4 groups (Q1 = 25th percentile, Q3 = 75th percentile)
- Quintiles: 5 groups

$$\text{Interquartile Range} = \text{Q}_3 - Q_1$$

3.2 Measures of Dispersion

LOS 3.b: Calculate, interpret, and evaluate measures of dispersion.

Range:

$$\text{Range} = \text{Max - Min}$$

Mean Absolute Deviation (MAD):

$$\text{MAD} = \sum |x_i - \bar{x}|_{\overline{n}}$$

Example: Returns: 30%, 12%, 25%, 20%, 23%, mean = 22%:

$$\text{MAD} = \frac{|8| + |10| + |3| + |2| + |1|}{5} = 4.8\%$$

Sample Variance:

$$\text{s}^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

Sample Standard Deviation:

$$\text{s} = \sqrt{s^2}$$

Example: Using same data: $s^2 = 44.5(\%)^2, s = 6.67\%$

Coefficient of Variation (CV):

$$CV = s_{\overline{\bar{x} \quad (Risk\ per\ unit\ of\ return)}}$$

Example: T-bills: mean 0.25%, s = 0.36% → CV = 1.44 Example: S&P 500: mean 1.09%, s = 7.30% → CV = 6.70 (higher risk per return)

Target Downside Deviation (Target Semideviation):

$$s_{target} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - B)^2}{n-1}}$$

(only for $x_i < B$*)*

Example: Target = 11%: Deviations: -6%, -18%, -9% → $s = \sqrt{(36 + 324 + 81)/5} = 9.39\%$

3.3 Skewness, Kurtosis, and Correlation

LOS 3.c: Interpret and evaluate measures of skewness and kurtosis.

Skewness:

- ***Positive Skew (Right-skewed):*** *Mean ¿ Median ¿ Mode. Long right tail. Large positive outliers pull mean up.*
- ***Negative Skew (Left-skewed):*** *Mean ¡ Median ¡ Mode. Long left tail. Large negative outliers pull mean down.*
- ***Symmetric:*** *Mean = Median = Mode.*

Skew pulls the mean. Mean is pulled toward the tail.

Kurtosis:

- ***Leptokurtic:*** *Excess kurtosis >0 (fat tails, more peaked). Higher probability of extreme outcomes → higher risk.*
- ***Platykurtic:*** *Excess kurtosis <0 (thin tails, less peaked).*
- ***Mesokurtic:*** *Excess kurtosis = 0 (normal distribution).*

$$\boxed{\text{Excess Kurtosis} = \text{Kurtosis - 3}}$$

LOS 3.d: Interpret correlation between two variables.

Correlation Coefficient:

$$\boxed{\rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho \leq +1}$$

Properties:

- *Measures* ***linear*** *relationship only.*
- $\rho = +1$: *Perfect positive correlation.*
- $\rho = -1$: *Perfect negative correlation.*
- $\rho = 0$: *No linear relationship.*
- *Correlation does* ***not*** *imply causation.*

- ***Spurious correlation*** *may result from chance or a third variable.*

Example: Cov = 0.0058, $\sigma_A = 0.0529, \sigma_B = 0.1114 : \rho = 0.0058/(0.05290.1114) = 0.9842$

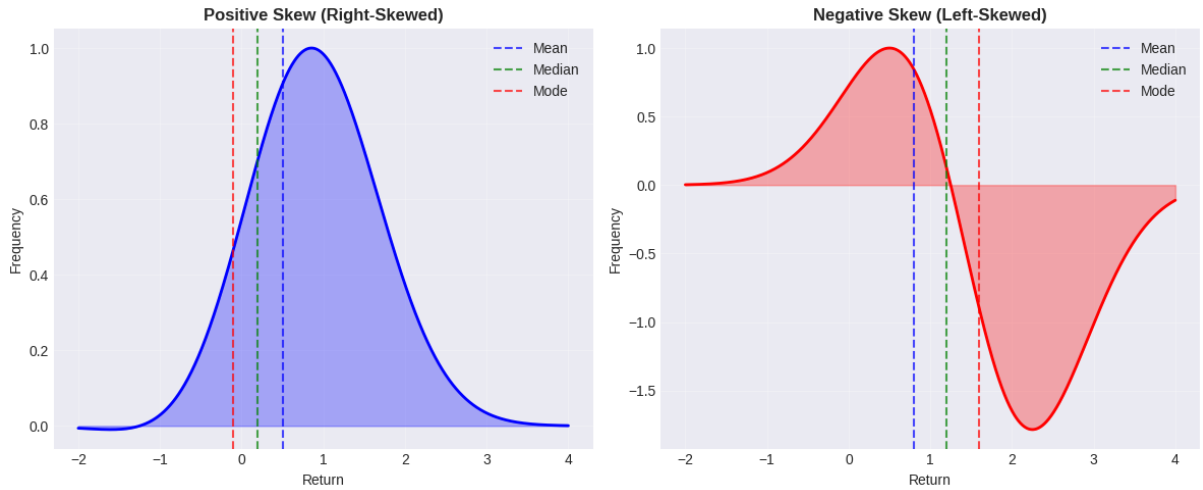

Figure 2: Skewness

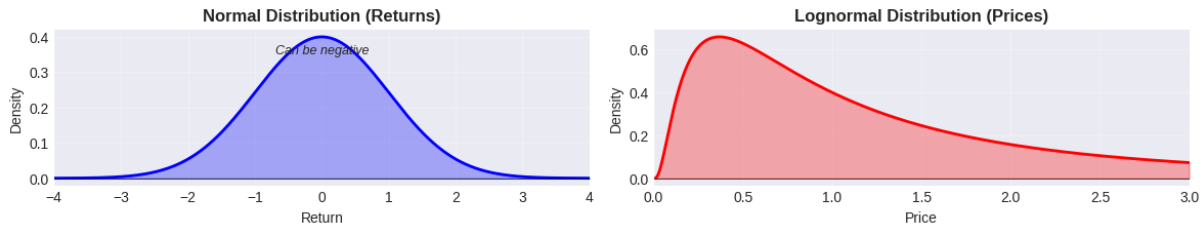

Figure 3: Normal vs Logarithm Distribution

4 Probability Trees and Conditional Expectations

LOS 4.a: Calculate expected values, variances, and standard deviations.

Expected Value:

$$\boxed{E(X) = \sum P(x_i) \cdot x_i}$$

Example: EPS: 10% prob \$1.80, 20% prob \$1.60, 40% prob \$1.20, 30% prob \$1.00:

$$E(EPS) = 0.10(1.80) + 0.20(1.60) + 0.40(1.20) + 0.30(1.00) = \$1.28$$

Variance (from probability model):

$$\boxed{Var(X) = \sum P(x_i) \cdot [x_i - E(X)]^2}$$

Standard Deviation:

$$\boxed{\sigma = \sqrt{Var(X)}}$$

Example: Stock returns: Boom(30%, 20%), Normal(50%, 12%), Slow(20%, 5%):

$$E(R) = 13\%, \quad Var = 0.0028, \quad \sigma = 5.29\%$$

LOS 4.b: Formulate an investment problem as a probability tree and explain conditional expectations.

Probability Tree: *Shows unconditional probabilities and conditional probabilities for subsequent events.*
Example: Economy: Good(60%), Poor(40%). If Good: EPS high(30% prob \$1.80) or low(70% prob \$1.70). If Poor: EPS high(60% prob \$1.30) or low(40% prob \$1.00).

$$E(EPS) = 0.18(1.80) + 0.42(1.70) + 0.24(1.30) + 0.16(1.00) = \$1.51$$

Conditional Expected Value: *Expected value given that some other event has occurred. Used to revise expectations when new information arrives.*

LOS 4.c: Calculate updated probabilities using Bayes' formula.

Bayes' Formula:

$$\boxed{P(A—B) = P(B—A)\frac{}{P(B)\times P(A)=\frac{P(B|A)P(A)}{P(B|A)P(A)+P(B|A^c)P(A^c)}}}$$

Example: P(Economy Outperform) = 60%. If Outperform: P(Stock Up) = 70%. If Underperform: P(Stock Up) = 20%. Given Stock Up, P(Outperform) = ?

$$P = \frac{0.70 \times 0.60}{(0.70 \times 0.60) + (0.20 \times 0.40)} = \frac{0.42}{0.42 + 0.08} = 84\%$$

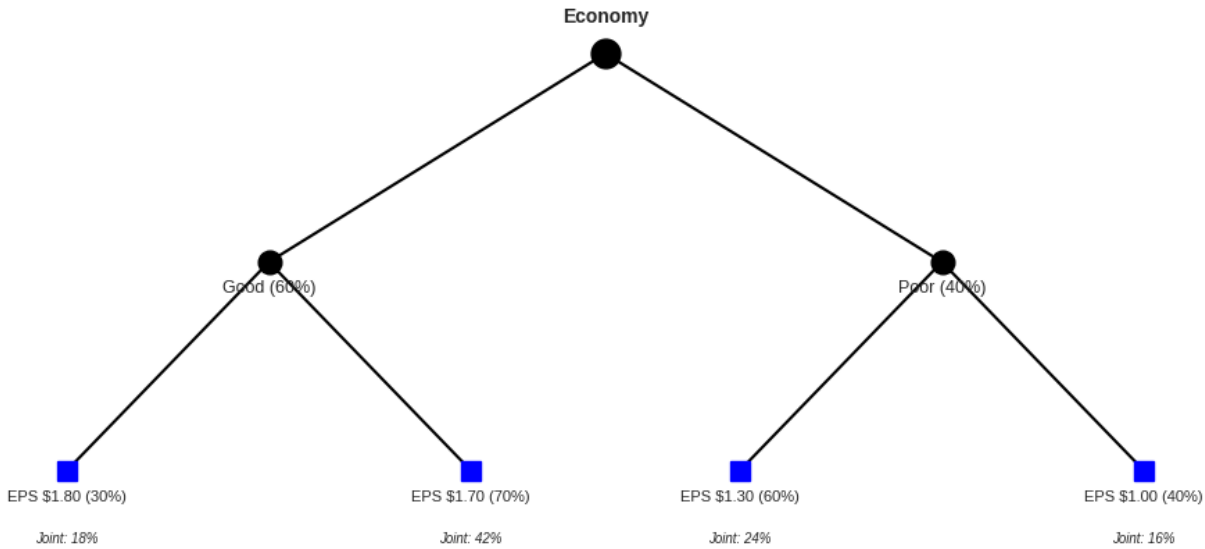

Figure 4: Probability Tree

5 Portfolio Mathematics

LOS 5.a: Calculate expected value, variance, standard deviation, covariances, and correlations of portfolio returns.

Expected Portfolio Return:

$$\boxed{E(R_p) = \sum w_i E(R_i)}$$

Portfolio Variance (2 assets):

$$\boxed{Var(R_p) = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B Cov_{AB}}$$

Covariance:

$$\boxed{Cov_{AB} = \rho_{AB}\sigma_A\sigma_B}$$

Correlation:

$$\boxed{\rho_{AB} = \frac{Cov_{AB}}{\sigma_A\sigma_B}}$$

Example: Cov = 0.0058, $_A = 0.0529, _B = 0.1114 := 0.0058/(0.05290.1114) = 0.9842$

Covariance Matrix: *Diagonal = variances, off-diagonal = covariances.*

LOS 5.b: Calculate covariance using a joint probability function.

$$\boxed{Cov(R_A, R_B) = \sum P(R_{Ai}, R_{Bi})[R_{Ai} - E(R_A)][R_{Bi} - E(R_B)]}$$

Example: Joint probabilities: (20%,30%) with 0.3, (12%,10%) with 0.5, (5%,0%) with 0.2:

$$E(R_A) = 13\%, E(R_B) = 14\%, Cov = 0.0058$$

LOS 5.c: Define shortfall risk, calculate safety-first ratio, and identify optimal portfolio using Roy's safety-first criterion.

Shortfall Risk: *Probability that portfolio return falls below threshold* R_L.

Roy's Safety-First Ratio:

$$SFR = E(R_p) - R_L \overline{\sigma_p}$$

Decision Rule: *Maximize SFR (minimize shortfall probability).*
Example: Target return = 3%, Portfolio A: E(R)=9%, =12% → SFR = (9-3)/12 = 0.5. Portfolio B: 11%,20% → 0.4. Portfolio C: 6.6%,8.2% → 0.44. Choose A.
Professor's Note: *If returns are normally distributed, the shortfall probability = F(-SFR) from z-table.*

6 Simulation Methods

LOS 6.a: Explain relationship between normal and lognormal distributions.

Key Relationship: *If x is normally distributed, then e^x is lognormally distributed.*

Why Lognormal for Asset Prices:

- *Prices cannot be negative (lognormal is bounded at zero).*
- *Continuously compounded returns are normally distributed.*
- $P_T = P_0 e^{r_{0,T}}$

Independently and Identically Distributed (IID):

- *Independent: Past returns don't predict future returns.*
- *Identically distributed: Mean and variance constant over time (stationarity).*

LOS 6.b: Describe Monte Carlo simulation.

Process:

1. *Specify probability distributions for risk factors.*
2. *Randomly generate values for each factor.*
3. *Value security using pricing model.*
4. *Repeat thousands of times.*
5. *Analyze distribution of values.*

Uses:

- *Valuing complex securities*
- *Value at Risk (VaR) estimation*
- *Testing scenarios outside historical range*

- *Portfolio stress testing*

Limitations:

- *Complex and computationally intensive*
- *Outputs only as good as assumptions*
- *Not an analytic method (can't provide insights like formulas)*

LOS 6.c: Describe bootstrap resampling.

Bootstrap Process:

1. *Start with original sample of size n.*
2. *Draw repeated samples of size n* ***with replacement****.*
3. *Calculate statistic for each sample.*
4. *Use distribution to estimate standard error.*

Advantages:

- *Can estimate distribution of complex statistics (e.g., median)*
- *No parametric assumptions needed*
- *Works when analytic methods don't exist*

Jackknife: *Calculate n sample means, each leaving out one observation. Less computationally demanding than bootstrap.*

7 Estimation and Inference

LOS 7.a: Compare and contrast sampling methods.

Method	***Description***
Simple Random	*Every item has equal probability of selection. Unbiased.*
Stratified Random	*Population divided into subgroups (strata), random samples from each proportional to stratum size.*
Cluster	*Random clusters selected, then all items (1-stage) or random items (2-stage) from clusters.*
Convenience	*Easily accessible data. Non-probability, high sampling error risk.*
Judgmental	*Researcher selects based on experience. Non-probability, bias risk.*

LOS 7.b: Explain the central limit theorem (CLT).

Central Limit Theorem: *For sample size* $n \geq 30$, *the sampling distribution of the sample mean* $\bar{x}$ *is approximately normal with:*

$$\mu_{\bar{x}} = \mu \quad \text{and} \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Standard Error of Sample Mean:

$$\boxed{s_{\bar{x}} = \frac{s}{\sqrt{n}}}$$

(when population unknown)

Example: $s = 20\%$, $n = 30$: $s_{\bar{x}} = 20\%/\sqrt{30} = 3.6\%$

8 Hypothesis Testing

LOS 8.a: Explain hypothesis testing components.

Definitions:

- ***Null Hypothesis (H_0):*** *What we want to reject. Always contains "=".*
- ***Alternative Hypothesis (H_a):*** *Concluded if H_0 is rejected.*
- ***Type I Error:*** *Rejecting true H_0 (probability = α, significance level).*
- ***Type II Error:*** *Failing to reject false H_0 (probability = β).*
- ***Power of Test:*** $1 - \beta$ *(probability of correctly rejecting false H_0).*
- ***p-value:*** *Smallest significance level at which H_0 can be rejected. If p-value ¡ , reject H_0.*

Decision Rule:

$$\text{Test Statistic} = \frac{\text{Sample Statistic} - \text{Hypothesized Value}}{\text{Standard Error}}$$

Reject H_0 if test statistic ¿ critical value (one-tailed) or —test statistic— ¿ critical value (two-tailed).

LOS 8.b: Construct hypothesis tests.

Test	***Statistic***	***df***
One population mean (large n, known)	$z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$	*N/A*
One population mean (unknown)	$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$	*n-1*
Difference in means (independent, equal variance)	$t = \frac{\bar{x}_1-\bar{x}_2}{s_p\sqrt{1/n_1+1/n_2}}$	*n1+n2-2*
Paired comparisons (dependent)	$t = \frac{\bar{d}}{s_d/\sqrt{n}}$	*n-1*

One population variance	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$	*n-1*
Two population variances	$F = \frac{s_1^2}{s_2^2}$ *(larger in numerator)*	*n1-1, n2-1*

LOS 8.c: Compare parametric and nonparametric tests.

- ***Parametric:*** *Assume specific distribution (e.g., normal). Examples: t-test, F-test, chi-square.*
- ***Nonparametric:*** *Fewer assumptions. Used when:*
  - *Distribution assumptions violated*
  - *Data are ranks (ordinal)*
  - *Testing non-parameter hypotheses (e.g., randomness)*
- *Examples: Spearman rank correlation, runs test.*

9 Parametric and Non-Parametric Tests of Independence

LOS 9.a: Explain parametric and nonparametric tests of correlation.

Test of Population Correlation = 0 (Parametric):

$$t = r\sqrt{n-2}\frac{}{\sqrt{1-r^2}}$$

with df = n-2

Example: r = 0.35, n = 42: t = 0.3540/(1-0.1225) = 2.363. Critical t(40, 0.05 two-tail) = 2.021 → Reject H0.

Spearman Rank Correlation (Nonparametric):

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

Test statistic same as parametric t-test for n ¿ 30.

LOS 9.b: Explain tests of independence based on contingency table data.

Chi-Square Test of Independence:

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij}-E_{ij})^2}{E_{ij}}$$

$$E_{ij} = \frac{(Row\ i\ total)\times(Column\ j\ total)}{Grand\ total}$$

$$df = (r\text{-}1)\times(c-1)$$

Example: 3×3 contingency table, ² = 27.47, df = 4, critical ²(0.05) = 9.488 → Reject independence.

10 Simple Linear Regression

LOS 10.a: Describe simple linear regression model.

Regression Equation:

$$Y_i = b_0 + b_1 X_i + \varepsilon_i$$

$$_i = \hat{b}_0 + \hat{b}_1 X_i$$

Least Squares Estimates:

$$_1 = \frac{Cov(X,Y)}{\sigma_X^2}, \quad \hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X}$$

Example: Cov = 0.000336, Var(X) = 0.000522 → b1 = 0.64. b0 = -4.05% - 0.64(-2.70%) = -2.3%

LOS 10.b: Explain assumptions underlying simple linear regression.

1. ***Linearity:*** *Relationship between X and Y is linear.*
2. ***Homoskedasticity:*** *Constant variance of errors (ε_i).*
3. ***Independence:*** *Errors are uncorrelated (no autocorrelation).*
4. ***Normality:*** *Errors are normally distributed (for valid hypothesis tests).*

Residual Plots:

- *Nonlinear pattern → linearity violated.*
- *Increasing spread → heteroskedasticity.*
- *Pattern over time → autocorrelation.*

LOS 10.c-d: Calculate and interpret measures of fit.

ANOVA Table:

Source	***df***	***SS***	***MS***

Regression	*1*	$SSR = (\hat{Y}_i - \bar{Y})^2$	$MSR = SSR/1$
Error	*n-2*	$SSE = (Y_i - \hat{Y}_i)^2$	$MSE = SSE/(n\text{-}2)$
Total	*n-1*	$SST = (Y_i - \bar{Y})^2$	

$$SST = SSR + SSE$$

Coefficient of Determination (R^2):

$$R^2 = \frac{SSR}{SST} = r^2$$

(for simple regression)

Standard Error of Estimate (SEE):

$$SEE = \sqrt{MSE} = \sqrt{\frac{SSE}{n-2}}$$

F-Test (Overall Significance):

$$F = MSR_{\overline{MSE = \frac{SSR/1}{SSE/(n-2)}}}$$

df = (1, n-2)

t-Test of Slope Coefficient:

$$t = {}_1 - b_{1\overline{s_{\hat{b}_1}}}$$

df = n-2

LOS 10.e: Calculate predicted value and prediction interval.

$$= {}_0 + \hat{b}_1 X_p$$

$$Prediction\ Interval = \ \pm t_c \times s_f$$

$$s_f^2 = SEE^2 \left[1 + \frac{1}{n} + \frac{(X_p - \bar{X})^2}{(n-1)s_X^2}\right]$$

LOS 10.f: Describe different functional forms.

Model	***Equation***	***Slope Interpretation***
Log-lin	$ln\ Y = b0 + b1\ X$	*Relative Y for absolute X*
Lin-log	$Y = b0 + b1\ ln\ X$	*Absolute Y for relative X*
Log-log	$ln\ Y = b0 + b1\ ln\ X$	*Relative Y for relative X (elasticity)*

11 Introduction to Big Data Techniques

LOS 11.a: Describe aspects of fintech relevant to financial data.

Fintech: *Technology applied to financial services. Includes:*

- *Handling large, varied datasets*
- *Artificial intelligence and machine learning*
- *Algorithmic trading*
- *Blockchain and cryptocurrencies*

LOS 11.b: Describe Big Data, AI, and machine learning.

Big Data Characteristics:

- ***Volume:*** *Terabytes to petabytes*
- ***Velocity:*** *Real-time (low latency) vs periodic (high latency)*
- ***Variety:*** *Structured (spreadsheets), semi-structured (photos), unstructured (video)*

Sources:

- *Traditional: Markets, financial reports, economic statistics*
- *Alternative: Social media, corporate exhaust, Internet of Things (IoT)*

Machine Learning:

- ***Supervised:*** *Labeled input and output data. Learns to map inputs to outputs.*
- ***Unsupervised:*** *Unlabeled data. Algorithm finds structure/patterns.*
- ***Deep Learning:*** *Multiple neural network layers. Image/speech recognition.*
- ***Risks:*** *Overfitting (too complex), underfitting (too simple), "black box" results.*

LOS 11.c: Describe applications to investment management.

- ***Text Analytics:*** *Analyzing unstructured text/voice data (earnings calls, filings).*
- ***Natural Language Processing (NLP):*** *Computers interpreting human language.*
- ***Algorithmic Trading:*** *Computerized trading based on rules. Includes high-frequency trading.*
- ***Risk Governance:*** *Simulations, stress testing, real-time risk monitoring.*

Part II

Economics

12 Firms and Market Structures

12.1 Breakeven, Shutdown, and Scale

LOS 12.a: Determine and interpret breakeven and shutdown points.

Breakeven:

$$P = ATC \quad or \quad TR = TC \quad (zero\ economic\ profit)$$

Short-Run Shutdown:

$$P < AVC \quad or \quad TR < TVC$$

Continue operating if P >AVC (cover some fixed costs). Shut down if P ¡ AVC.

Long-Run Shutdown:

$$P < ATC \quad or \quad TR < TC$$

Exit market if expected to persist.

Economies and Diseconomies of Scale:

- ***Economies of Scale:*** *LRATC decreases as output increases. Causes: specialization, mass production, better equipment.*
- ***Diseconomies of Scale:*** *LRATC increases as output increases. Causes: bureaucracy, motivation problems.*
- ***Minimum Efficient Scale:*** *Lowest point on LRATC curve.*

12.2 Characteristics of Market Structures

LOS 12.b: Describe characteristics of market structures.

	Perfect	***Monopolistic***	***Oligopoly***	***Monopoly***
Number of firms	*Many*	*Many*	*Few*	*One*
Barriers to entry	*Very low*	*Low*	*High*	*Very high*

Product differentiation	*None (homogeneous)*	*High*	*Some/None*	*No substitutes*
Pricing power	*None*	*Some*	*Significant*	*Significant*
Non-price competition	*None*	*Heavy (advertising)*	*Heavy*	*Some*
Demand curve	*Horizontal (elastic)*	*Downward-sloping*	*Downward-sloping*	*Downward-sloping (steep)*
Long-run profit	*Zero*	*Zero*	*Positive possible*	*Positive possible*

12.3 Monopolistic Competition

LOS 12.c: Explain supply and demand under monopolistic competition.

Short-Run:

- *Downward-sloping demand (MR <P)*
- *Profit-maximizing: MR = MC*
- *Can earn positive economic profits*

Long-Run:

- *Entry of new firms shifts demand down*
- *Zero economic profit: P = ATC*
- *But P >MC (inefficient, excess capacity)*

12.4 Oligopoly

LOS 12.d: Explain supply and demand under oligopoly.

Key Feature: *Interdependence among firms.*

Kinked Demand Curve:

- *Competitors match price cuts but not price increases*
- *Demand more elastic above kink (price increases lose market share)*

- *Demand less elastic below kink (price cuts matched)*
- *Gap in MR curve → prices sticky*

Cournot Model: *Simultaneous quantity decisions. Each firm assumes competitor's quantity fixed. Long-run: equal market share.*

Stackelberg Model: *Sequential decisions. First mover (leader) gets larger market share and higher profit.*

Nash Equilibrium: *No firm can unilaterally change price/quantity to improve outcome.*

Collusion/Cartel: *Agreement to restrict output and raise price. Often illegal. More successful with: fewer firms, similar products, similar costs, frequent purchases, certain retaliation.*

Dominant Firm Model: *One large firm (low costs) sets price. Competitive fringe takes that price.*

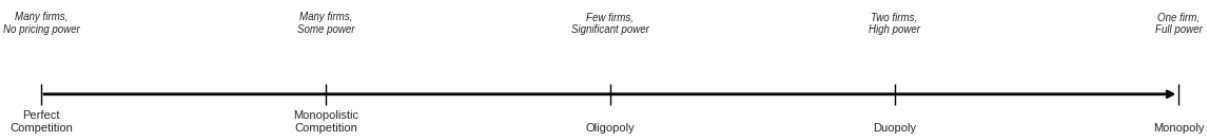

Figure 5: Market Strcture

12.5 Identifying Market Structures

LOS 12.e: Identify market structure and describe concentration measures.

N-Firm Concentration Ratio: *Sum of market shares of top N firms. Simple but insensitive to mergers.*

Herfindahl-Hirschman Index (HHI):

$$\boxed{HHI = \sum_{i=1}^{N} s_i^2}$$

More sensitive to mergers. Range: 0 to 10,000.
Example: Firms: 25%, 15%, 15%, 10%, 5%, rest. 4-firm ratio = 65%. HHI = 625+225+225+100 = 1175. After merger of 25% and 15%: new 4-firm ratio = 70%, new HHI = 1600+225+100+25 = 1950.

Limitations: *Neither measure accounts for barriers to entry or potential competition.*

13 Understanding Business Cycles

LOS 13.a: Describe the business cycle and its phases.

Four Phases:

1. ***Expansion:*** *Real GDP increasing. Employment, consumption, investment rising. Inflation accelerating.*
2. ***Peak:*** *GDP stops increasing. Growth rates slow. Inflation high.*
3. ***Contraction (Recession):*** *GDP decreasing. Employment falling. Inflation decreasing with lag.*
4. ***Trough:*** *GDP stops decreasing. Recovery begins. Inflation moderate.*

Alternative Measures:

- ***Classical Cycle:*** *Based on real GDP level.*
- ***Growth Cycle:*** *GDP relative to trend.*
- ***Growth Rate Cycle:*** *GDP growth rate (preferred by practitioners).*

LOS 13.b: Describe credit cycles.

Credit Cycle: *Cyclical fluctuations in interest rates and credit availability.*

- *Expansion: Loose credit, lower rates, potential asset bubbles.*
- *Contraction: Tight credit, higher rates.*
- *Amplifies business cycles.*
- *Typically longer duration than business cycles.*

LOS 13.c: Describe how resource use, consumer activity, housing, and external trade vary.

Inventory-Sales Ratio:

- *Increases late in expansion (unplanned buildup as sales slow)*

- *Decreases near trough (depletion as sales accelerate)*
- *Firms adjust production to restore normal ratio*

Labor: *Firms adjust hours/overtime first, then hiring/layoffs (costly to adjust workforce).*

Physical Capital: *Use existing capital more/less intensively. Invest in new capacity only if expansion likely to persist.*

Consumer Spending:

- *Durable goods: Highly cyclical (postpone during uncertainty)*
- *Services: Somewhat cyclical*
- *Nondurable goods: Least cyclical*

Housing: *Affected by: mortgage rates, income/housing cost ratio, speculation, demographics (25-40 age group).*

External Trade:

- *Imports: Rise with domestic GDP, domestic currency appreciation*
- *Exports: Rise with trading partner GDP, domestic currency depreciation*

Economic Indicators:

- ***Leading:*** *Turn before cycle (stock market, building permits)*
- ***Coincident:*** *Turn with cycle (GDP, employment, industrial production)*
- ***Lagging:*** *Turn after cycle (unemployment, inventory-sales ratio)*

14 Fiscal Policy

LOS 14.a: Compare monetary and fiscal policy.

	Monetary Policy	***Fiscal Policy***
Authority	*Central bank*	*Government (legislative/executive)*
Tools	*Policy rate, reserve requirements, open market operations*	*Taxes, government spending*
Primary goal	*Price stability*	*Economic growth, employment*
Implementation speed	*Faster*	*Slower (political process)*

LOS 14.b: Describe roles and objectives of fiscal policy.

Objectives:

1. *Influence aggregate demand (stabilization)*
2. *Redistribute wealth/income*
3. *Allocate resources among industries*

Automatic Stabilizers: *Built-in mechanisms that automatically counter cyclical movements:*

- *Recession: Tax receipts fall, unemployment benefits rise → deficit increases (expansionary)*
- *Boom: Tax receipts rise, social spending falls → deficit decreases (contractionary)*

Debate on National Debt:

- ***Concerned:*** *Higher future taxes (disincentives), crowding out, loss of confidence.*
- ***Not concerned:*** *Ricardian equivalence (private savings offset deficits), productive capital investment, domestic ownership of debt.*

Crowding-Out Effect: *Government borrowing increases interest rates, reducing private investment.*

Ricardian Equivalence: *Taxpayers increase savings in anticipation of future taxes, offsetting deficit spending. No effect on aggregate demand.*

LOS 14.c: Describe tools of fiscal policy.

Spending Tools:

- *Transfer payments (entitlements): Social Security, unemployment insurance*
- *Current spending: Government purchases of goods/services*
- *Capital spending: Infrastructure (roads, schools, hospitals)*

Revenue Tools:

- *Direct taxes: Income, wealth, corporate, capital gains*
- *Indirect taxes: Sales, VAT, excise (can discourage consumption)*

Desirable Tax Attributes: *Simplicity, efficiency (minimize market interference), fairness (horizontal and vertical equity), sufficiency.*

Fiscal Multiplier:

$$\boxed{Fiscal\ Multiplier = 1\frac{}{1-MPC(1-t)}}$$

Example: MPC = 0.8, t = 0.25: Multiplier = 1/(1 - 0.8×0.75) = 1/(1 - 0.6) = 2.5

Balanced Budget Multiplier: *Equal increase in G and T increases aggregate demand (positive multiplier).*

LOS 14.d: Explain implementation and difficulties.

Lags:

1. ***Recognition lag:*** *Time to identify economic problem.*
2. ***Action lag:*** *Time to enact policy changes (political process).*
3. ***Impact lag:*** *Time for policy to affect economy.*

Challenges:

- *Misreading economic statistics*
- *Crowding out*
- *Supply shortages (if problem is supply, not demand)*
- *Deficit limits*
- *Multiple targets (can't address unemployment and inflation simultaneously)*

Structural Budget Deficit: *Cyclically adjusted deficit (what deficit would be at full employment). Better indicator of policy stance than actual deficit.*

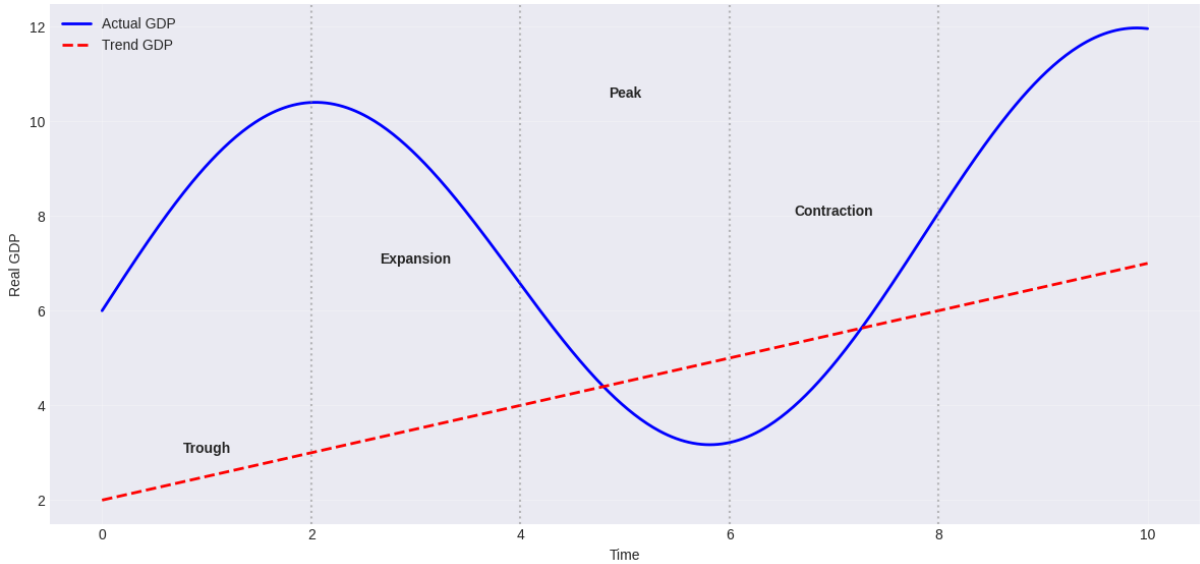

Figure 6: Business Cycle

15 Monetary Policy

LOS 15.a: Describe roles and objectives of central banks.

Roles:

1. *Sole supplier of currency (fiat money)*
2. *Banker to government and other banks*
3. *Regulator and supervisor of payments system*
4. *Lender of last resort (prevents bank runs)*
5. *Holder of gold and foreign exchange reserves*
6. *Conductor of monetary policy*

Primary Objective: *Price stability (inflation target 2-3%). Not zero (risk of deflation).*

Secondary Objectives:

- *Exchange rate stability*
- *Full employment*
- *Sustainable economic growth*
- *Moderate long-term interest rates*

LOS 15.b: Describe tools of monetary policy and transmission mechanism.

Tools:

1. ***Policy Rate:*** *Discount rate (US), refinancing rate (ECB), repo rate (UK).*
2. ***Reserve Requirements:*** *Higher requirement → less lending → tighter money.*
3. ***Open Market Operations:*** *Buy securities → increase money supply → lower rates. Sell securities → decrease money supply → higher rates.*

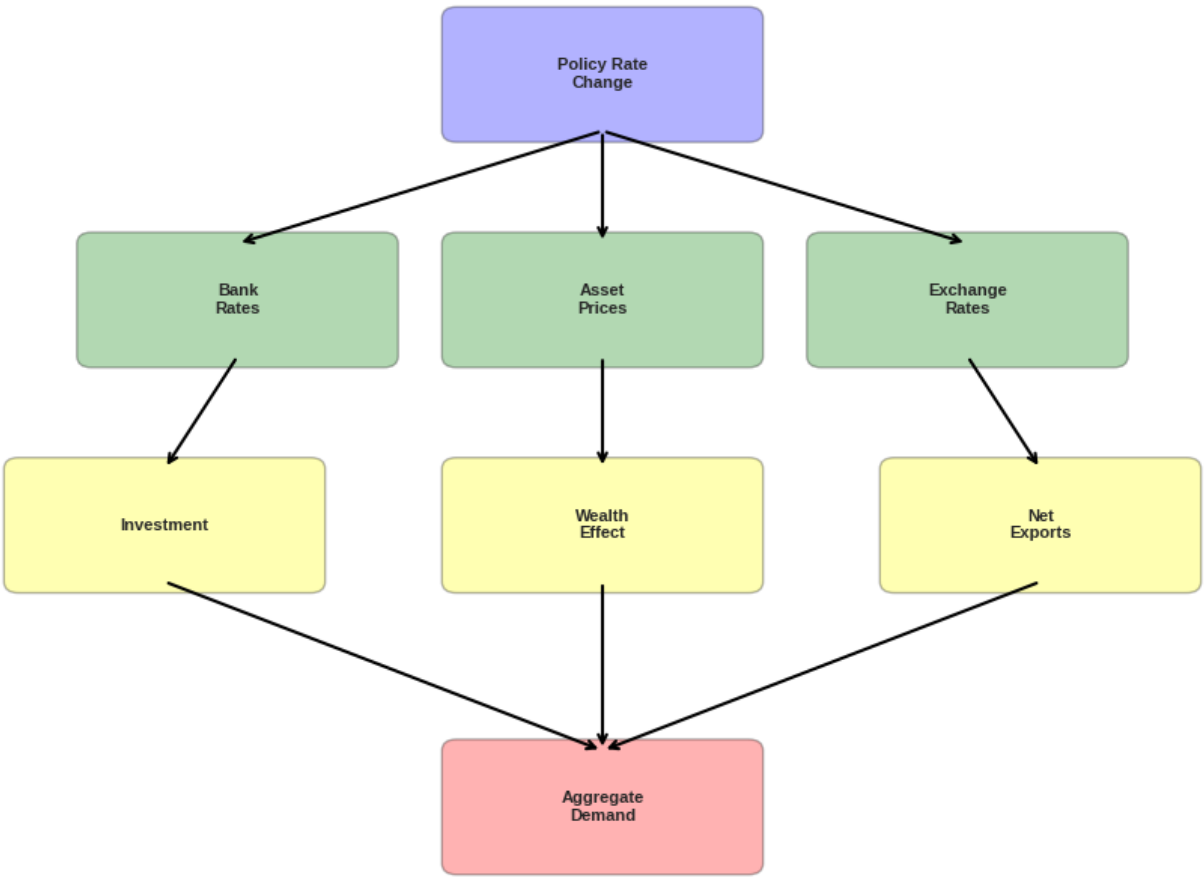

Figure 7: Monetary Transmission Mechanism

Transmission Mechanism (Contractionary Policy Example):

1. *Increase policy rate*
2. *Bank lending rates increase → reduce consumption/investment*
3. *Asset prices decrease → wealth effect → reduce consumption*
4. *Expectations decrease → reduce spending*
5. *Currency appreciates → reduce exports*
6. *Aggregate demand decreases → inflation decreases*

Effects of Expansionary Policy: *Lower rates, currency depreciates, higher asset prices, increased consumption/investment, higher GDP, higher inflation.*

LOS 15.c: Describe qualities of effective central banks.

Three Qualities:

1. ***Independence:*** *Free from political interference.Operational (set policy rate) and target (set inflation goal).*

2. ***Credibility:*** *Follows through on stated intentions.Self-fulfilling prophecies.*

3. ***Transparency:*** *Discloses economic indicators and policy reasoning.*

Inflation Targeting: *Most common. Neutral rate = real trend growth + inflation target. Policy rate above neutral = contractionary.*

Exchange Rate Targeting: *Peg currency to another (often USD). Must match inflation rate of target country. Can run out of reserves.*

Limitations of Monetary Policy:

- *Long-term rates may move opposite to policy rates (inflation expectations)*
- *Liquidity trap (highly elastic money demand)*
- *Zero lower bound (can't go below 0%) → deflation difficult to reverse*
- *Banks may not lend even with excess reserves*

Quantitative Easing (QE): *Large-scale purchases of longer-term securities when policy rate near zero. Aim to reduce long-term rates and increase lending.*

LOS 15.d: Explain interaction of monetary and fiscal policy.

Monetary	***Fiscal***	***GDP***	***Interest Rates***
Expansionary	*Expansionary*	*Strong increase*	*Mixed*
Contractionary	*Contractionary*	*Strong decrease*	*Mixed*
Expansionary	*Contractionary*	*Mixed*	*Lower*
Contractionary	*Expansionary*	*Mixed*	*Higher*

16 Introduction to Geopolitics

LOS 16.a: Describe geopolitics from cooperation vs competition perspective.

Geopolitics: *Interactions among nations (state actors: governments) and nonstate actors (corporations, NGOs, individuals). Study of how geography affects these interactions.*

Cooperation Spectrum: Countries lie on spectrum from cooperative to noncooperative on various issues (diplomacy, military, economics, culture).

National Interests: *Countries cooperate when it advances their interests. Hierarchy: survival first, then economic prosperity, then other goals.*

Soft Power: *Ability to influence without force (cultural exchange, institutions).*

LOS 16.b: Describe geopolitics and its relationship with globalization.

Globalization: *Long-term trend toward worldwide integration of economic activity and cultures. Trade as % of GDP increased from 25% (1970s) to 60% (pre-2008).*

Nationalism: *Pursuing own economic interests independently of other countries (opposition to globalization).*

Four Archetypes:

- ***Autarky:*** *Noncooperation + Nationalism (self-reliance, state-dominated)*
- ***Hegemony:*** *Noncooperation + Globalization (large enough to influence others)*
- ***Bilateralism:*** *Cooperation + Nationalism (cooperates one-on-one, not multilateral)*
- ***Multilateralism:*** *Cooperation + Globalization (extensive international cooperation)*

LOS 16.c: Describe functions of international organizations.

International Monetary Fund (IMF):

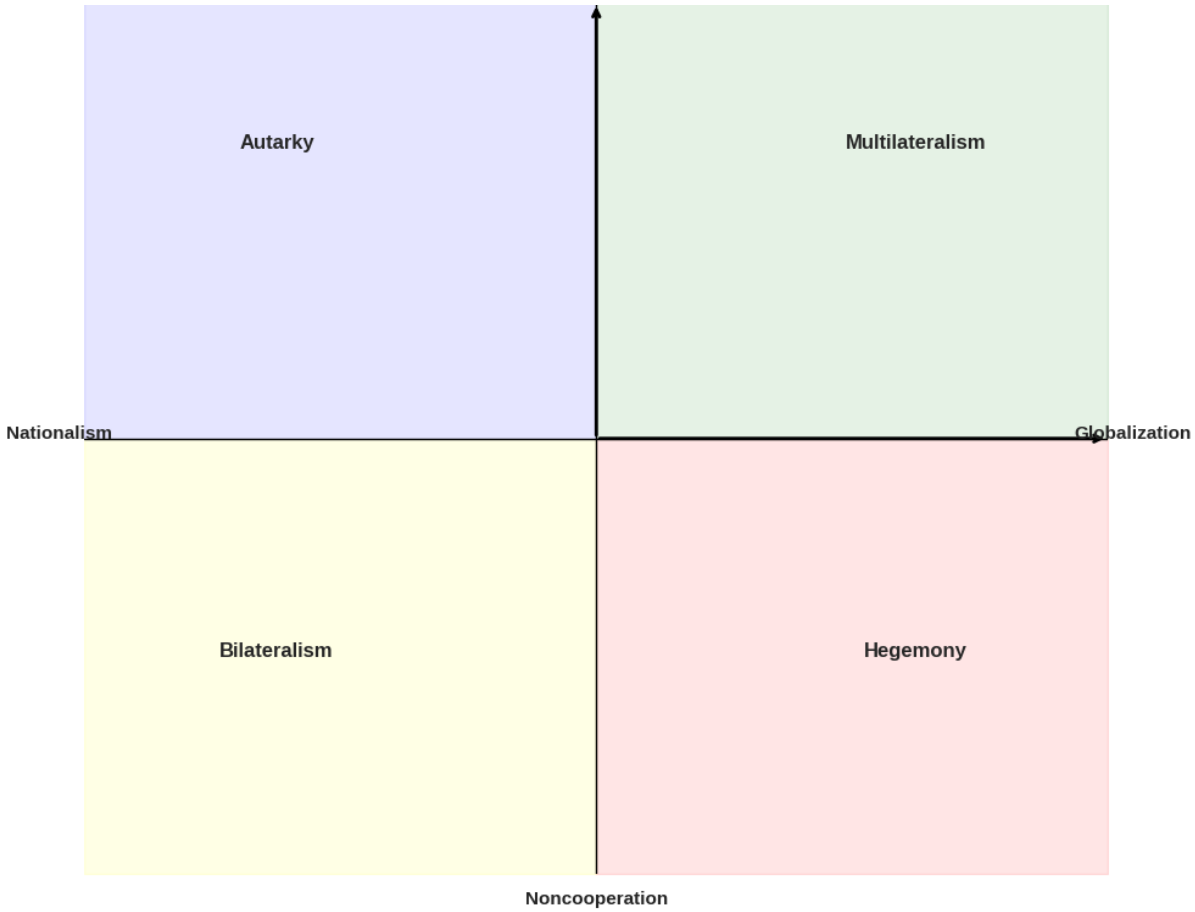

Figure 8: Tariff Effects

- *Promote international monetary cooperation*
- *Facilitate balanced growth of trade*
- *Promote exchange rate stability*
- *Assist with balance of payments difficulties*

World Bank:

- *Financial and technical assistance to developing countries*
- *Fight poverty*
- *IBRD (middle-income) and IDA (poorest countries)*
- *Low-interest loans, interest-free credits, grants*

World Trade Organization (WTO):

- *Global rules of trade between nations*
- *Ensure trade flows smoothly, predictably, freely*
- *Dispute settlement process*
- *Multilateral trade agreements*

LOS 16.d: Describe geopolitical risk.

Categories:

- ***Event Risk:*** *Known timing, unknown outcome (elections)*
- ***Exogenous Risk:*** *Unanticipated events (war, rebellion)*
- ***Thematic Risk:*** *Known factors with long-term effects (migration, cyber)*

Analysis Dimensions:

- *Probability (likelihood)*
- *Impact (effect on investment outcomes)*
- *Velocity (how quickly effects occur)*

Black Swan Risk: *Low-probability exogenous events with substantial short-term effects.*

LOS 16.e: Describe tools of geopolitics.

National Security Tools: *Armed conflict, espionage, bilateral/multilateral security agreements.*

Economic Tools:

- *Cooperative: Free trade areas, common markets, economic unions*
- *Noncooperative: Domestic content requirements, voluntary export restraints, nationalization*

Financial Tools: *Foreign investment, currency exchange, sanctions.*

LOS 16.f: Describe impact on investments.

Analysis Focus: *Focus on risks with high likely impact. Determine if impacts are discrete (company/industry) or broad (country/region/world).*

Scenario Analysis: *Qualitative or quantitative. Identify signposts (data signaling increasing/decreasing likelihood).*

17 International Trade

LOS 17.a: Describe benefits and costs of international trade.

Benefits:

- *Comparative advantage (specialization increases total output)*
- *Economies of scale*
- *Greater variety of goods*
- *Increased competition (reduces monopoly power)*
- *More efficient resource allocation*

Costs:

- *Job losses in import-competing industries*
- *Increased income inequality*
- *Short-run adjustment costs*

Key Insight: *Gains from trade exceed losses overall (winners could compensate losers and still be better off).*

LOS 17.b: Compare types of trade restrictions.

Tariffs: *Taxes on imports. Increase domestic price, decrease quantity imported, increase domestic production, generate government revenue. Deadweight loss.*

Quotas: *Quantity limits on imports. Same price effect as tariff. Quota rents (difference between world price and domestic price times quantity) may go to foreign exporters if licenses not sold.*

Voluntary Export Restraint (VER): *Exporter voluntarily limits exports. Similar to quota but rents go to foreign exporters. Greater welfare loss to importing country.*

Export Subsidies: *Government payments to exporters. Benefit exporters, harm domestic consumers (higher prices), benefit foreign consumers (lower world price if large exporter).*

Minimum Domestic Content: *Requirement that some percentage of product content be domestic.*

Effects Summary (Importing Country): *All restrictions: reduce imports, increase domestic price, decrease consumer surplus, increase producer surplus, create deadweight loss.*

LOS 17.c: Explain motivations for trading blocs.

Type	***Features***
Free Trade Area	*No internal barriers to goods/services*
Customs Union	*Free trade area + common external tariff*
Common Market	*Customs union + free movement of labor/capital*
Economic Union	*Common market + common economic policies/institutions*
Monetary Union	*Economic union + single currency*

Examples: *NAFTA/USMCA (free trade area), EU (economic union), Eurozone (monetary union).*

Trade Creation vs Trade Diversion:

- *Trade creation: Shift from high-cost domestic to low-cost member (beneficial)*
- *Trade diversion: Shift from low-cost nonmember to higher-cost member (harmful)*

Figure 9: Tariff Effects

18 Capital Flows and the FX Market

LOS 18.a: Describe the foreign exchange market.

FX Market Participants:

- ***Sell Side:*** *Large multinational banks (primary dealers, originators of forwards)*
- ***Buy Side:*** *Corporations, investment accounts (real money and leveraged), governments, central banks, retail*

Hedging vs Speculating:

- *Hedging: Reducing existing FX risk*
- *Speculating: Increasing FX risk (taking positions)*

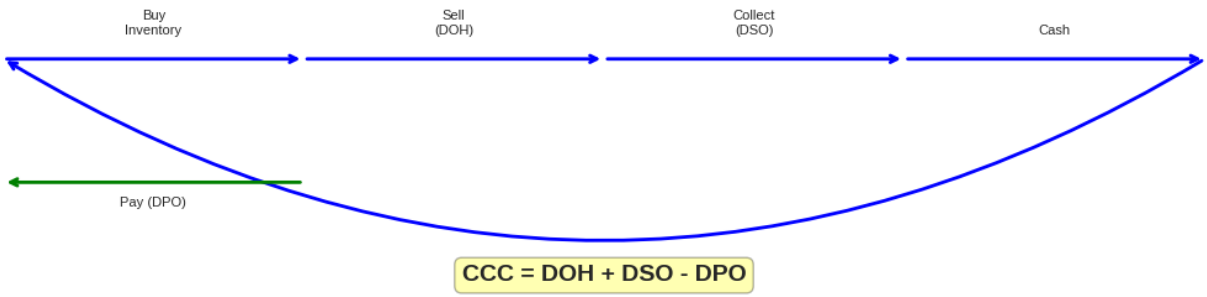

Figure 10: Cash Conversion Cycle

Exchange Rate Notation:

$$USD/EUR = 1.25 \quad (1.25 \text{ USD per 1 EUR})$$

- *USD = price currency (numerator)*
- *EUR = base currency (denominator)*

Nominal vs Real Exchange Rates:

$$Real = Nominal \times \frac{CPI_{base}}{CPI_{price}}$$

Example: Nominal = 1.60 USD/GBP, $CPI_US = 110, CPI_UK = 112$: *Real* $= 1.60(112/110) = 1.629$

Percentage Change in Currency: *Example: USD/EUR from 1.42 to 1.39: EUR depreciated 1.39/1.42 - 1 = -2.11%. USD appreciated: (1/1.39)/(1/1.42) - 1 = 2.16% (not symmetric).*

LOS 18.b: Describe exchange rate regimes.

No Own Currency:

- *Formal dollarization: Use another country's currency*
- *Monetary union: Common currency (e.g., Eurozone)*

Own Currency:

- *Currency board: Fixed exchange rate, fully backed by foreign reserves (Hong Kong)*
- *Conventional fixed peg: ±1% bands*
- *Target zone: Wider bands (±2%)*
- *Crawling peg: Periodic adjustments (active or passive)*
- *Crawling bands: Bands widen over time*
- *Managed floating: No specific target, intervention to influence*
- *Independently floating: Market determined*

Trade and Capital Flows:

$$X - M = (S - I) + (T - G)$$

Trade deficit = private sector deficit + government deficit. Capital flows offset trade imbalances.

LOS 18.c: Describe objectives of capital restrictions.

- *Reduce volatility of domestic asset prices*
- *Maintain fixed exchange rates*
- *Keep domestic interest rates low (prevent capital outflows)*
- *Protect strategic industries (national security)*

19 Exchange Rate Calculations

LOS 19.a: Calculate currency cross-rates.

Cross Rate Calculation:

$$A_{\overline{C=\frac{A}{B}\times\frac{B}{C}}}$$

Example: MXN/USD = 10.70, USD/AUD = 0.60 → MXN/AUD = 10.70 × 0.60 = 6.42 Example: CHF/USD = 1.7799, NZD/USD = 2.2529 → CHF/NZD = 1.7799/2.2529 = 0.7900

LOS 19.b: Explain arbitrage relationship between spot and forward rates.

Interest Rate Parity (IRP):

$$F_{\overline{S=\frac{1+i_{price}}{1+i_{base}}}}$$

Example: S = 4.5671 ABE/DUB, $i_ABE = 5\%, i_DUB = 3\% : F = 4.5671(1.05/1.03) = 4.6558$

Forward Points: *Points = last decimal place. Spot 1.4158, +25.3 points → forward = 1.4158 + 0.00253 = 1.41833*

Forward Premium/Discount (Base Currency):

$$Premium/Discount = F_{\overline{S-1}}$$

Example: Spot = 1.312, Forward = 1.320: Premium = 1.320/1.312 - 1 = 0.610% (euro at forward premium)

Arbitrage Opportunity: *If IRP violated, borrow low-interest currency, convert, invest, sell forward, repay loan, keep profit.*

Part III

Corporate Issuers

20 Organizational Forms, Features, and Ownership

LOS 20.a: Compare organizational forms of businesses.

	Sole Proprietorship	***Partnership***	***Corporation***
Legal entity	*No*	*No*	*Yes*
Liability	*Unlimited*	*Unlimited (general partners)*	*Limited*
Taxation	*Personal*	*Personal*	*Double (corporate + dividend)*
Access to capital	*Low*	*Medium*	*High*
Separation of ownership/management	*No*	*No (general partners)*	*Yes*
Transfer of ownership	*Difficult*	*Difficult*	*Easy (if public)*

Limited Partnership: *General partners (operate, unlimited liability) + limited partners (invest only, limited liability).*

Limited Liability Partnership (LLP): *All partners have limited liability (professional services only in US).*

LOS 20.b: Describe key features of corporate issuers.

- *Separate legal entity (articles of incorporation)*
- *Shareholders have voting rights (elect board of directors)*
- *Limited liability (can't lose more than invested)*
- *Board hires managers, declares dividends*
- *Double taxation: Corporate tax on earnings + personal tax on dividends*

Example: Double Taxation: Pretax \$100, corporate tax 25% → \$90 after tax, dividend tax 15% → \$13.50 tax, total tax \$23.50 (23.5%). If 40% paid

out: corporate tax \$25, dividend tax = 40%×75×0.15 = \$4.50, total tax \$29.50 (29.5%).

LOS 20.c: Compare publicly and privately owned corporate issuers.

	Public Company	***Private Company***
Share trading	*Exchange-listed, transparent*	*No exchange, illiquid*
Disclosure requirements	*Extensive (quarterly/annual reports)*	*Limited*
Free float	*Actively traded shares*	*N/A*
Regulatory burden	*High*	*Low*
Time horizon	*Short-term pressure*	*Longer-term focus*

Going Public Methods:

1. *IPO: Issues new shares, raises capital, uses underwriter*
2. *Direct listing: Lists existing shares, no new capital, faster/cheaper*
3. *SPAC acquisition: Blank check company raises capital, then acquires*

21 Investors and Other Stakeholders

LOS 21.a: Compare financial claims and motivations of lenders and shareholders.

	Lenders (Debtholders)	***Shareholders***
Priority of claims	*Higher (before equity)*	*Residual (after debt)*
Return	*Fixed (interest)*	*Variable (dividends + capital gains)*
Upside potential	*Limited (promised payments)*	*Unlimited*
Risk	*Lower*	*Higher*
Cost of capital	*Lower (less risky)*	*Higher*
Motivation	*Receive promised payments, protect against default*	*Maximize firm value*

Conflict: *Shareholders may prefer higher risk (higher potential return). Debtholders prefer lower risk. Covenants in debt agreements protect debtholders.*
Example: Leverage Effect: \$1,000 revenue, \$800 expenses, \$1,000 assets. 100% equity: ROE = 200/1000 = 20%. 50/50 debt/equity at 10%: interest = 50, net = 150, ROE = 150/500 = 30%. But if revenue drops 15%: 100% equity ROE = 50/1000 = 5%, 50/50 ROE = 0/500 = 0%.

LOS 21.b: Describe stakeholder groups and compare their interests.

Stakeholder Theory vs Shareholder Theory:

- *Shareholder theory: Primary focus is maximizing shareholder value.*
- *Stakeholder theory: Balance interests of all stakeholders.*

Key Stakeholders:

- ***Shareholders:*** *Maximize firm value, voting rights*
- ***Lenders:*** *Receive promised payments, covenants protect them*

- ***Board of Directors:*** *Protect shareholder interests. Inside vs independent directors. One-tier vs two-tier boards.*
- ***Senior Managers:*** *Compensation (salary, bonus, perks), job security*
- ***Employees:*** *Wages, working conditions, job security. ESOPs align interests.*
- ***Customers:*** *Quality products at fair prices*
- ***Suppliers:*** *Ongoing relationship, timely payment*
- ***Government:*** *Tax revenue, regulatory compliance*

LOS 21.c: Describe environmental, social, and governance (ESG) factors.

Environmental Factors: *Climate change, air/water pollution, deforestation, energy efficiency, waste management, water scarcity. Physical risk (weather) and transition risk (regulations). Stranded assets.*

Social Factors: *Customer privacy, information security, customer satisfaction, employee engagement, diversity and inclusion, labor relations, community relations.*

Governance Factors: *Board composition, executive compensation, internal audit, bribery and corruption, political contributions, lobbying.*

Why ESG Matters:

1. *Government regulations increasing*
2. *Material impact on financial results*
3. *Younger investors demand it*

22 Corporate Governance

LOS 22.a: Describe principal-agent conflicts.

Principal-Agent Relationship: *Principal (owner) hires agent (manager) to act in principal's interest. Conflict arises when interests don't align.*

Agency Costs: *Costs of principal-agent conflict. Direct (monitoring) and indirect (lost opportunities).*

Common Conflicts:

- *Managers put in insufficient effort*
- *Risk preferences: Managers more risk-averse (their job at stake)*
- *Empire building: Compensation tied to size*
- *Entrenchment: Avoid risk, mimic competitors*
- *Self-dealing: Exploit firm resources for personal benefit*
- *Controlling vs minority shareholders: Dual-class structures*
- *Shareholders vs creditors: Shareholders prefer higher risk*

LOS 22.b: Describe corporate governance mechanisms.

Shareholder Mechanisms:

- *Annual general meeting (AGM): Vote on directors, auditor, other matters*
- *Proxy voting: Assign voting rights to another person*
- *Activist shareholders: Proxy contests, shareholder lawsuits, tender offers*
- *Hostile takeovers: Discipline poor management*

Creditor Mechanisms:

- *Bond indenture: Legal document specifying rights and obligations*

- *Covenants: Restrictions on borrower actions*
- *Collateral: Specific assets pledged*
- *Creditor committees: Protect interests in financial distress*

Board Mechanisms:

- *Audit committee: Oversee financial reporting, internal controls, external auditor*
- *Nominating/governance committee: Board elections, corporate governance code*
- *Compensation committee: Executive pay (should be independent directors)*
- *Risk committee: Risk policy and tolerance (financial services)*

Employee Mechanisms: *Labor laws, unions, ESOPs, board representation (some countries).*

LOS 22.c: Describe risks and benefits.

Risks of Poor Governance:

- *Weak controls, accounting fraud*
- *Self-dealing, related-party transactions*
- *Legal and reputational risk*
- *Default risk*
- *Lower firm value*

Benefits of Effective Governance:

- *Improved operational efficiency*
- *Lower cost of debt*
- *Better financial performance*
- *Higher firm value*
- *Reduced default risk*

23 Working Capital and Liquidity

LOS 23.a: Explain the cash conversion cycle (CCC).

Cash Conversion Cycle:

$$CCC = DOH + DSO - DPO$$

- *DOH (Days Inventory on Hand): Average days to sell inventory*
- *DSO (Days Sales Outstanding): Average days to collect receivables*
- *DPO (Days Payables Outstanding): Average days to pay suppliers*

Interpretation: *Lower CCC is better (less capital tied up in working capital).*

Cost of Trade Credit:

$$EAR = \left(1 + \frac{d}{1-d}\right)^{365/(c-b)} - 1$$

For terms "2/10 net 30": EAR = (1 + 0.02/0.98)$^{365/20} - 1 = 44.6\%$
Example: If bank rate is 8%, borrow from bank to take discount (much cheaper).

LOS 23.b: Explain liquidity and compare liquidity levels.

Primary Liquidity Sources: *Cash, marketable securities, bank borrowings, operating cash flow.*

Secondary Liquidity Sources: *Suspending dividends, delaying capex, selling assets, issuing equity, debt restructuring, bankruptcy protection. Send negative signals, higher cost.*

Drags on Liquidity: *Slow inflows: high DOH (excess/obsolete inventory), high DSO (slow collections).*

Pulls on Liquidity: *Accelerated outflows: low DPO (suppliers demanding faster payment).*

Liquidity Ratios:

$$Current\ Ratio = CA_{\overline{CL}}$$

$$Quick\ Ratio = Cash + MS + AR_{\overline{CL}}$$

$$\boxed{Cash\ Ratio = Cash + MS_{\overline{CL}}}$$

LOS 23.c: Describe methods for managing working capital.

Conservative Approach: *High current assets, long-term financing. Lower risk, lower return.*

Aggressive Approach: *Low current assets, short-term financing. Higher risk, higher return.*

Moderate Approach: *Permanent current assets funded with long-term capital. Seasonal current assets funded with short-term.*

24 Capital Investments and Capital Allocation

LOS 24.a: Describe types of capital investments.

- ***Going Concern:*** *Maintain business (replace equipment) or reduce costs (efficiency)*
- ***Regulatory/Compliance:*** *Required by law (often zero revenue)*
- ***Expansion:*** *Grow business (new markets, new products)*
- ***Other:*** *New lines of business, startups, acquisitions*

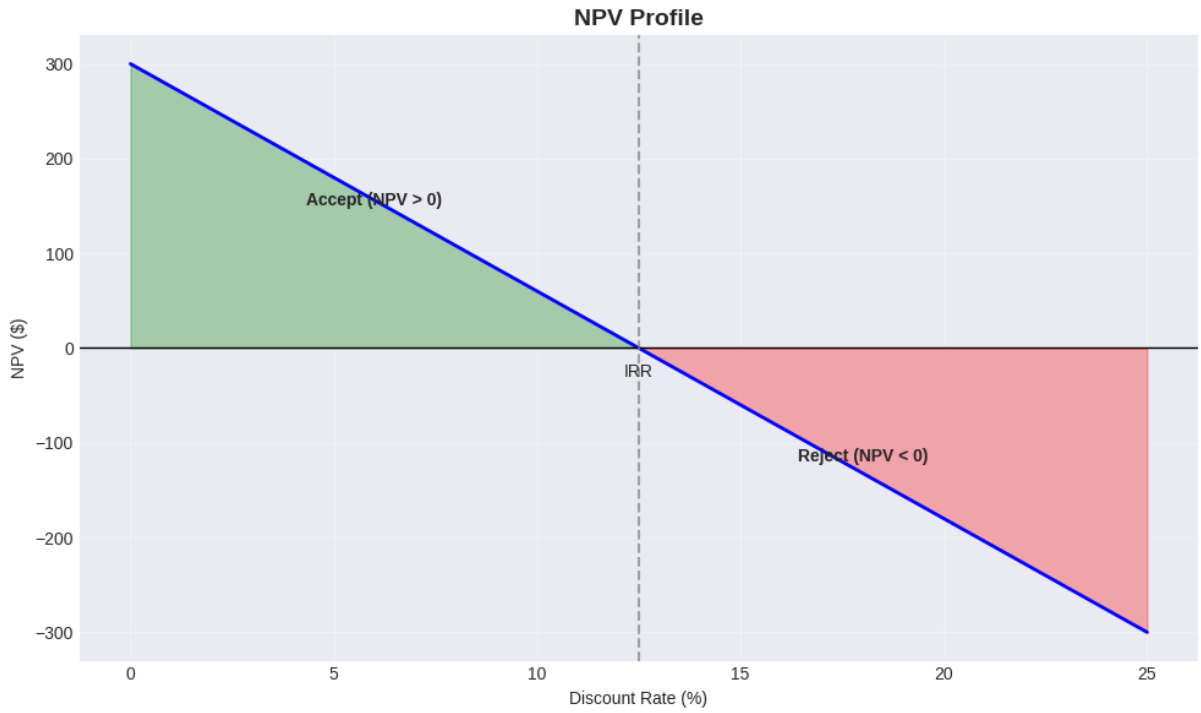

Figure 11: NPV Profile

LOS 24.b: Calculate NPV, IRR, and ROIC.

Net Present Value (NPV):

$$NPV = \sum_{t=0}^{n} \frac{CF_t}{(1+r)^t}$$

Decision: Accept if NPV >0 (increases shareholder wealth).

Internal Rate of Return (IRR):

$$\sum_{t=0}^{n} \frac{CF_t}{(1+IRR)^t} = 0$$

Decision: Accept if IRR >required return.

NPV vs IRR:

- *NPV: Direct measure of value added, assumes reinvestment at cost of capital.*
- *IRR: Percentage return, margin of safety, assumes reinvestment at IRR (unrealistic).*
- *For unconventional cash flows (multiple sign changes), IRR may have multiple solutions.*

Return on Invested Capital (ROIC):

$$\boxed{ROIC = NOPAT_{\overline{Average\ Invested\ Capital = Operating\ Margin \times Capital\ Turnover}}}$$

Compare to required return. If ROIC ¿ required return, firm adds value.

LOS 24.c: Describe principles and pitfalls.

Principles:

1. *Use after-tax cash flows (not accounting income)*
2. *Include only incremental cash flows*
3. *Ignore sunk costs (already incurred, not affected by decision)*
4. *Include spillover effects (cannibalization, externalities)*
5. *Account for timing (TVM)*

Cognitive Errors: *Poor forecasting, ignoring cost of internal funds, incorrect inflation adjustment.*

Behavioral Biases: *Pet projects of senior management, static capital budgets, focusing on short-term EPS/ROE, failure to generate alternatives.*

LOS 24.d: Describe types of real options.

- ***Timing Options:*** *Delay investment until more information available*
- ***Abandonment Options:*** *Exit project if future cash flows insufficient*

- ***Expansion/Growth Options:*** *Make additional investments if project succeeds*
- ***Flexibility Options:*** *Price-setting or production-flexibility (overtime, different inputs)*
- ***Fundamental Options:*** *Project value depends on underlying asset price (copper mine)*

Real options add value. Never negative (option not exercised if unfavorable).

25 Capital Structure

LOS 25.a: Calculate weighted-average cost of capital (WACC).

$$WACC = w_d \times r_d \times (1 - T) + w_e \times r_e$$

Example: 50% debt at 8%, 50% equity at 11%, tax 30%: WACC = 0.5×0.08×0.7 + 0.5×0.11 = 0.028 + 0.055 = 8.3%

LOS 25.b: Explain factors affecting capital structure.

Factors Increasing Debt Capacity:

- *Stable/predictable revenue and cash flow*
- *Low business risk*
- *Tangible, liquid collateral*
- *Low existing debt*
- *High corporate tax rate (debt tax shield more valuable)*

Life Cycle:

- *Start-up: Equity only (no cash flow, no collateral)*
- *Growth: Some debt (increasing cash flow)*
- *Mature: Significant debt (stable cash flow)*

LOS 25.c: Explain Modigliani-Miller propositions.

MM I (No Taxes): *Capital structure irrelevant. Firm value independent of debt/equity mix.*

MM II (No Taxes):

$$r_e = r_0 + (r_0 - r_d)\frac{D}{E}$$

Cost of equity increases with leverage, WACC constant.

MM I (With Taxes):

$$V_L = V_U + t \times D$$

Firm value increases with debt (tax shield). Optimal = 100% debt.

MM II (With Taxes):

$$\boxed{r_e = r_0 + (r_0 - r_d)(1 - t)\frac{D}{E}}$$

WACC decreases with leverage.

Static Tradeoff Theory: *Optimal capital structure balances tax shield benefit vs expected costs of financial distress. Costs include direct (legal fees) and indirect (lost opportunities, loss of trust).*

LOS 25.d: Describe optimal and target capital structures.

Pecking Order Theory: *Based on asymmetric information. Preference order:*

1. *Internal funds (retained earnings)*
2. *Debt*
3. *External equity (least preferred, signals overvaluation)*

Free Cash Flow Hypothesis: *Debt disciplines managers (less free cash flow to waste).*

Target Capital Structure: *What firm aims to achieve over time. Fluctuates due to market conditions and financing opportunities.*

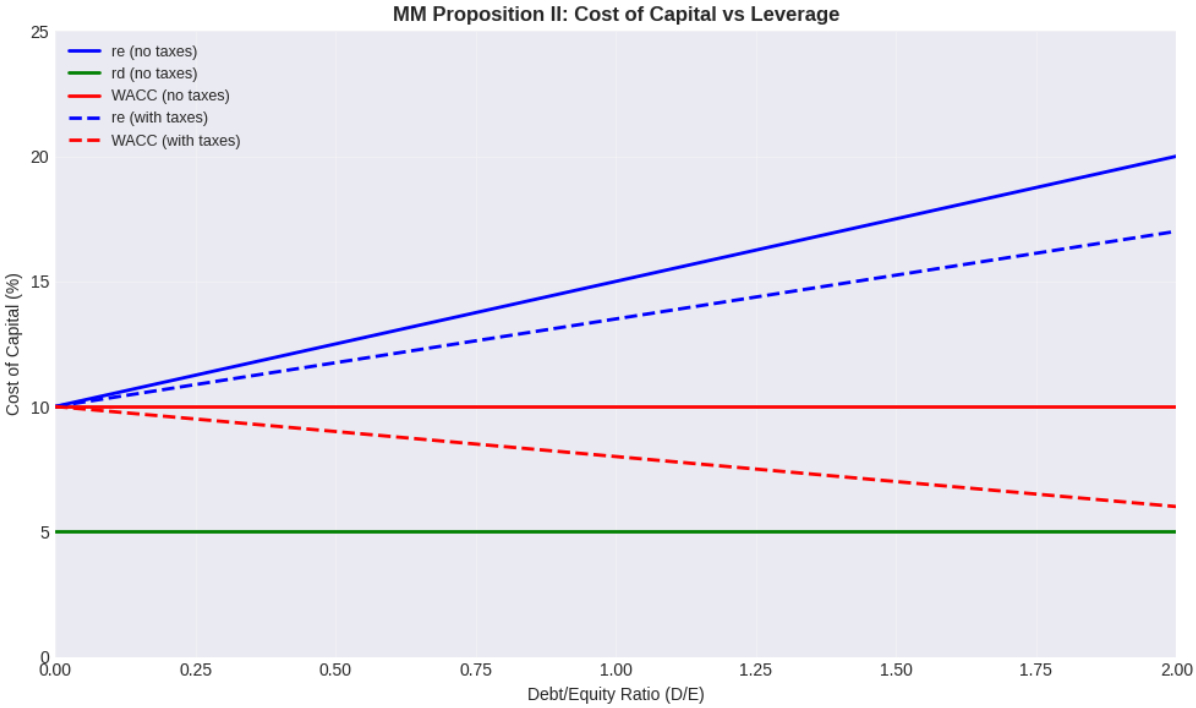

Figure 12: Tariff Effects

26 Business Models

LOS 26.a: Describe key features of business models.

Five Questions a Business Model Answers:

1. ***Who:*** *Target customers, segmentation, acquisition cost*
2. ***What:*** *Product/service, how it meets needs, differentiation*
3. ***How:*** *Key assets, key suppliers, value chain*
4. ***Where:*** *Channel strategy (direct, intermediaries, omnichannel). B2B vs B2C.*
5. ***How much:*** *Pricing strategy*

Pricing Strategies:

- *Price discrimination: Tiered, dynamic, value-based, auction*
- *Bundling: Multiple products together*
- *Razors-and-blades: Low margin on equipment, high margin on consumables*
- *Add-on: Options after purchase decision*
- *Penetration pricing: Low price to gain market share*
- *Freemium: Basic free, premium features paid*
- *Hidden revenue: Free to user, revenue from ads/data*
- *Subscription: Recurring payments*
- *Licensing/Franchising: Others sell your product for a fee*

Value Proposition: *How customers will value the product given competing products and prices.*

Value Chain (Porter's Five Activities): *Inbound logistics, operations, outbound logistics, marketing, sales and service.*

LOS 26.b: Describe various types of business models.

Conventional Models: *Natural resource producers, manufacturers, distributors, retailers, banks, brokers, service providers.*

Other Models:

- *Private label/contract manufacturers (Kirkland at Costco)*
- *Licensing agreements (Marvel characters on lunch boxes)*
- *Value-added resellers (installation, service, customization)*
- *Software as a Service (SaaS)*
- *Ultra-low-cost airlines*
- *Discount brokers*

Network Effects: *Value increases as user base grows (WhatsApp, Facebook, eBay). Two-sided networks (Airbnb: hosts and guests).*

Crowdsourcing: *User contributions (Wikipedia content, Waze traffic, open-source software).*

Annexure: Numerical Graphs

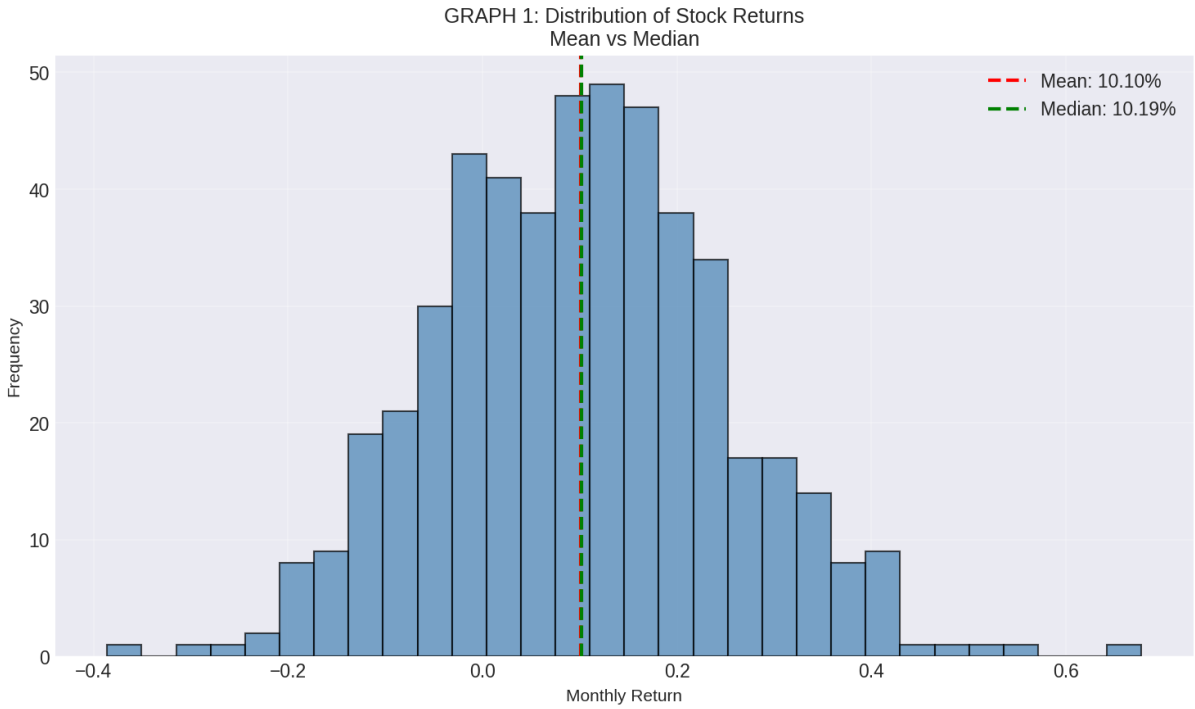

Figure 13: Distribution of Stock Returns - Mean vs Median

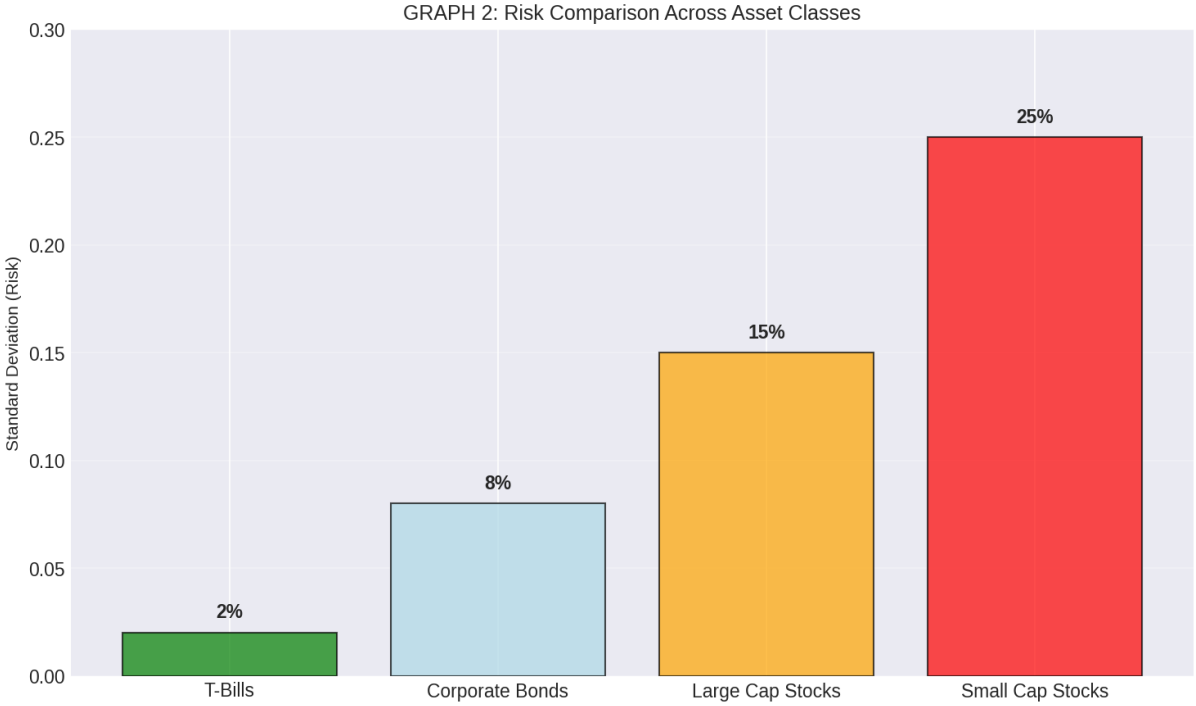

Figure 14: Risk Comparison Across Asset Classes

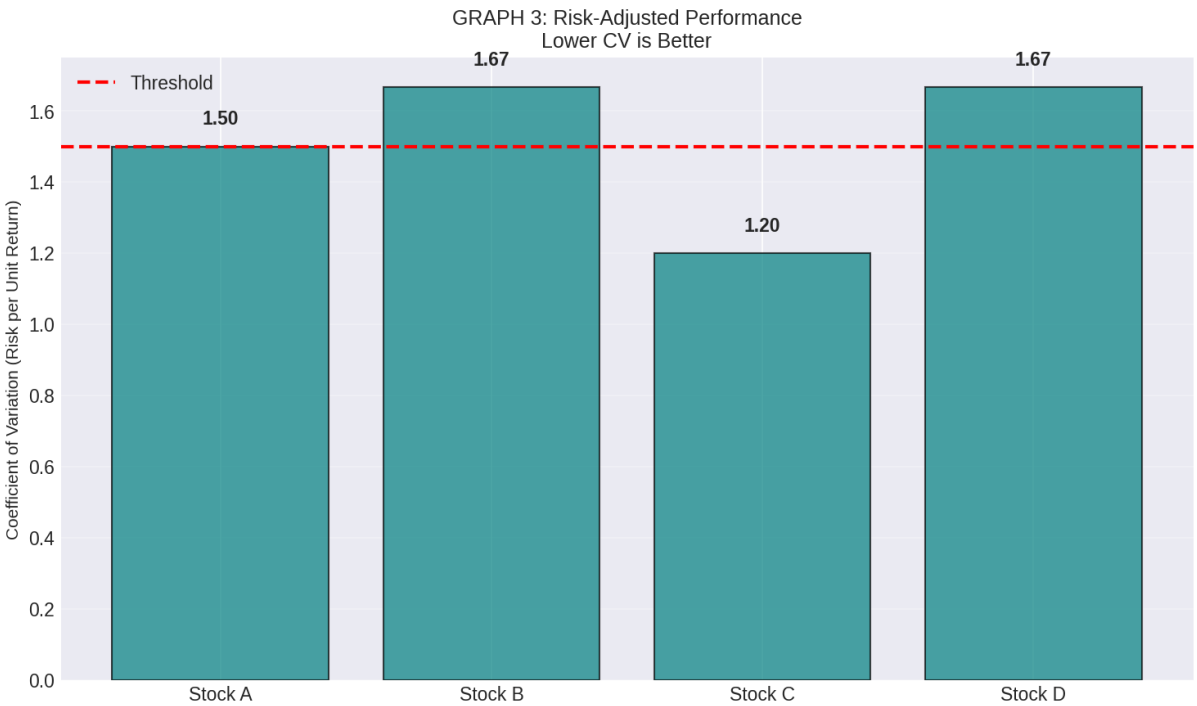

Figure 15: Coefficient of Variation (Risk per Unit Return)

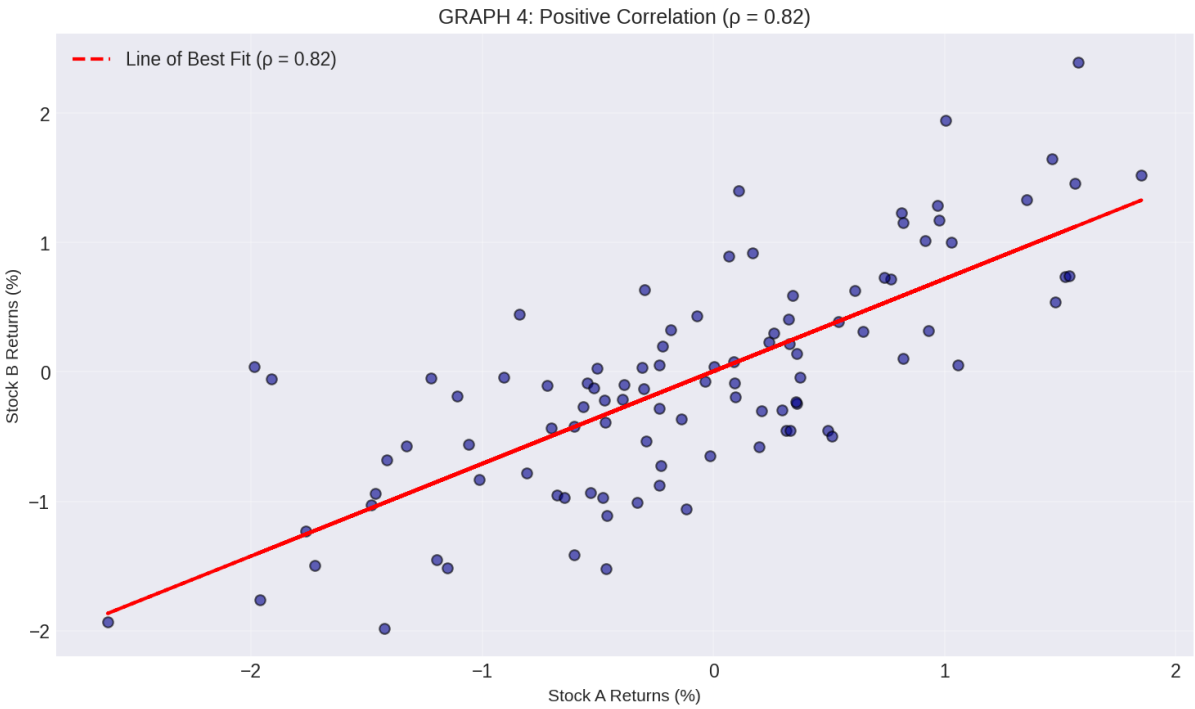

Figure 16: Positive Correlation Scatter Plot

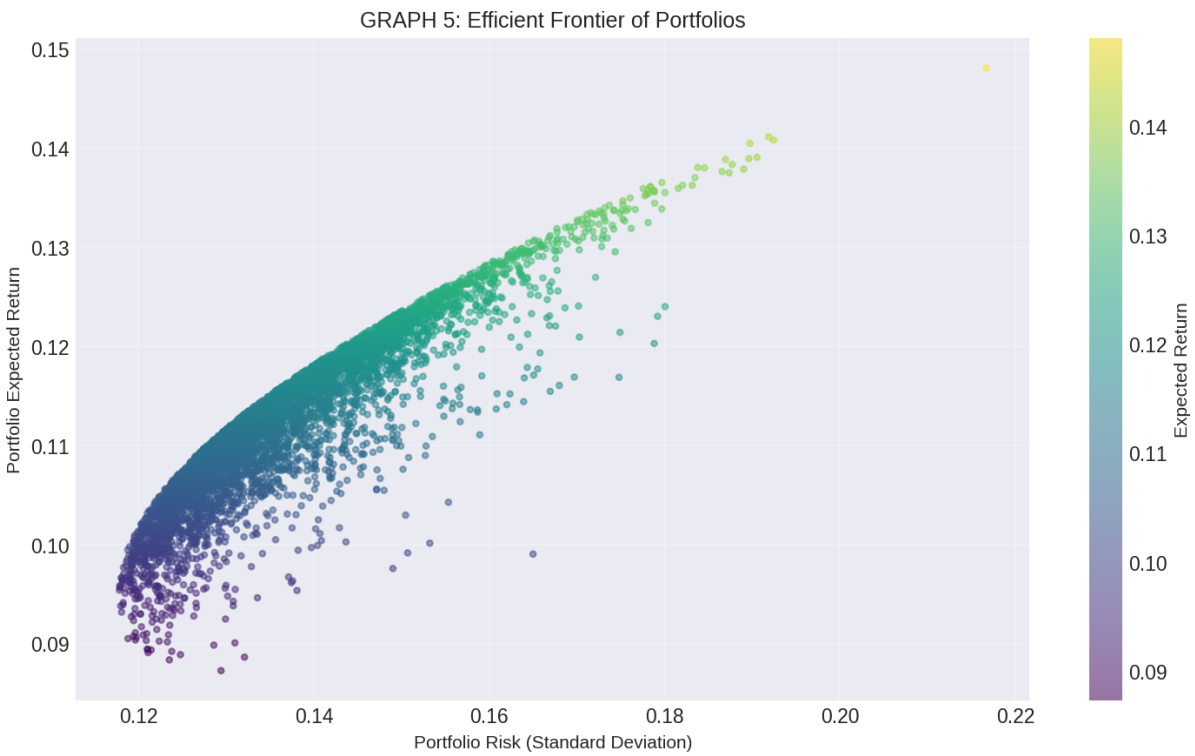

Figure 17: Portfolio Efficient Frontier

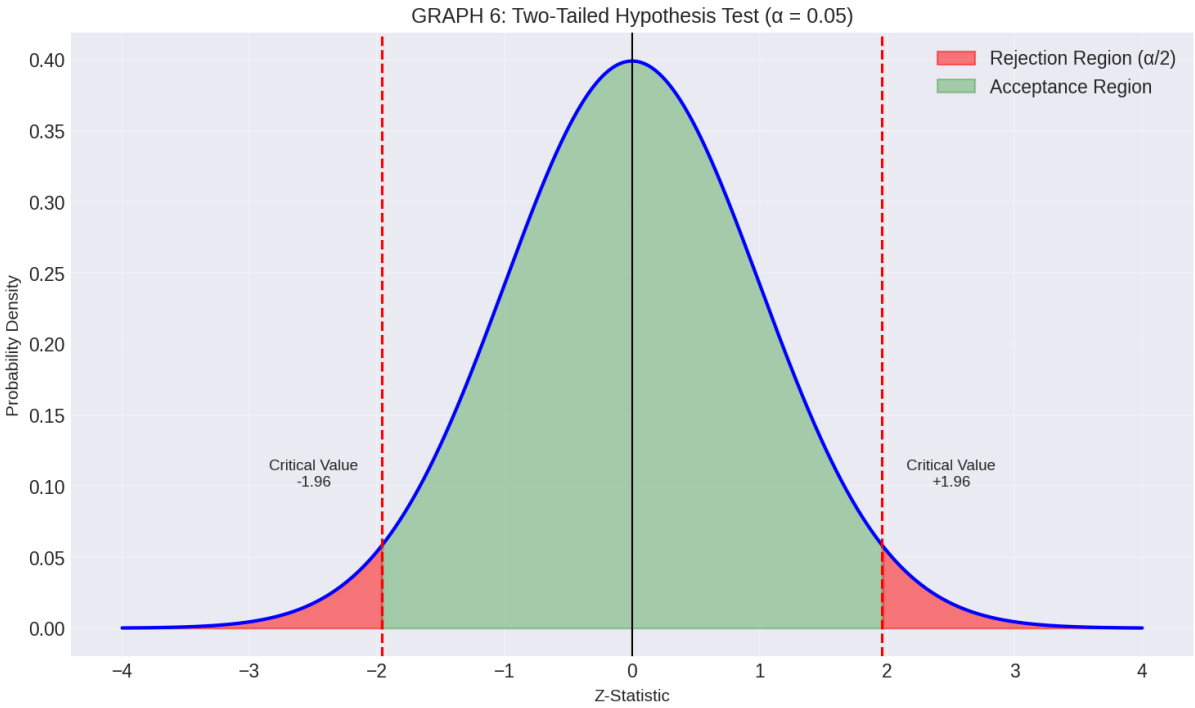

Figure 18: Two-Tailed Hypothesis Test (Critical Regions)

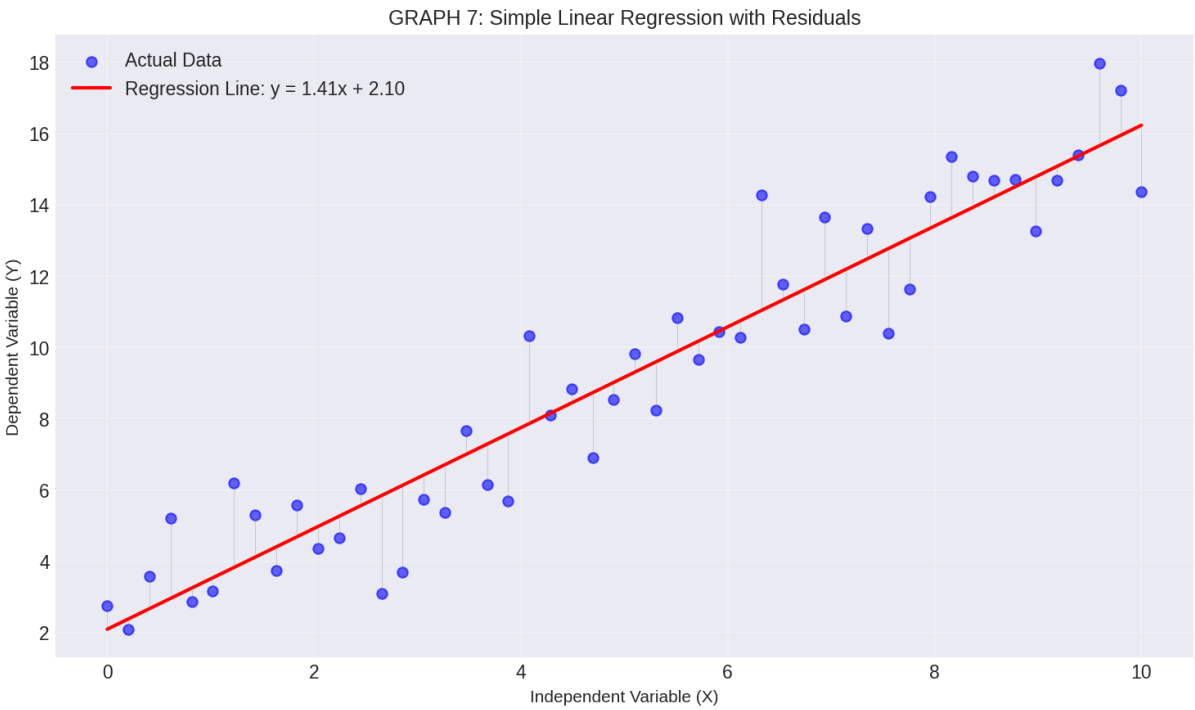

Figure 19: Simple Linear Regression with Residuals

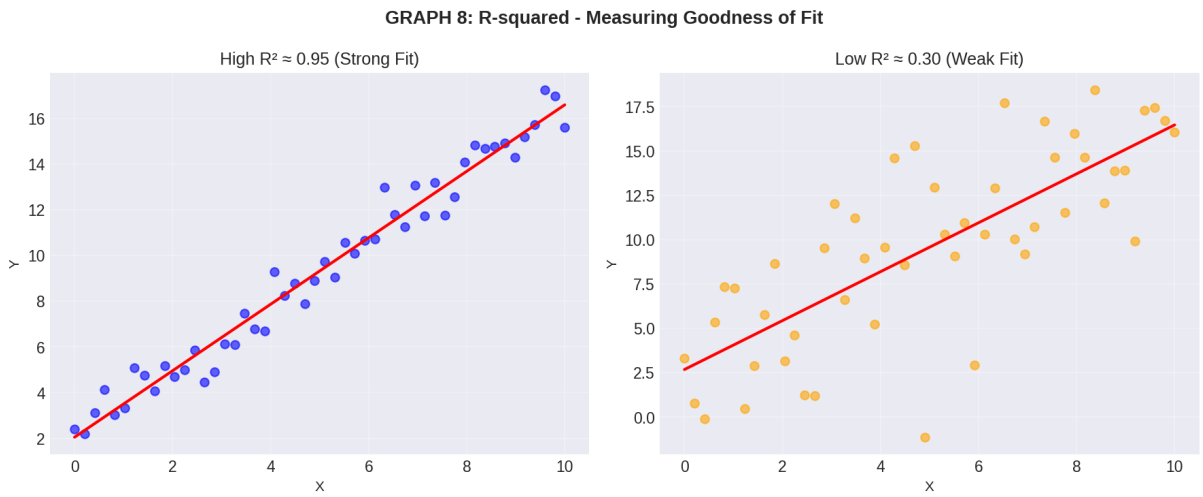

Figure 20: R-squared: High vs Low Goodness of Fit

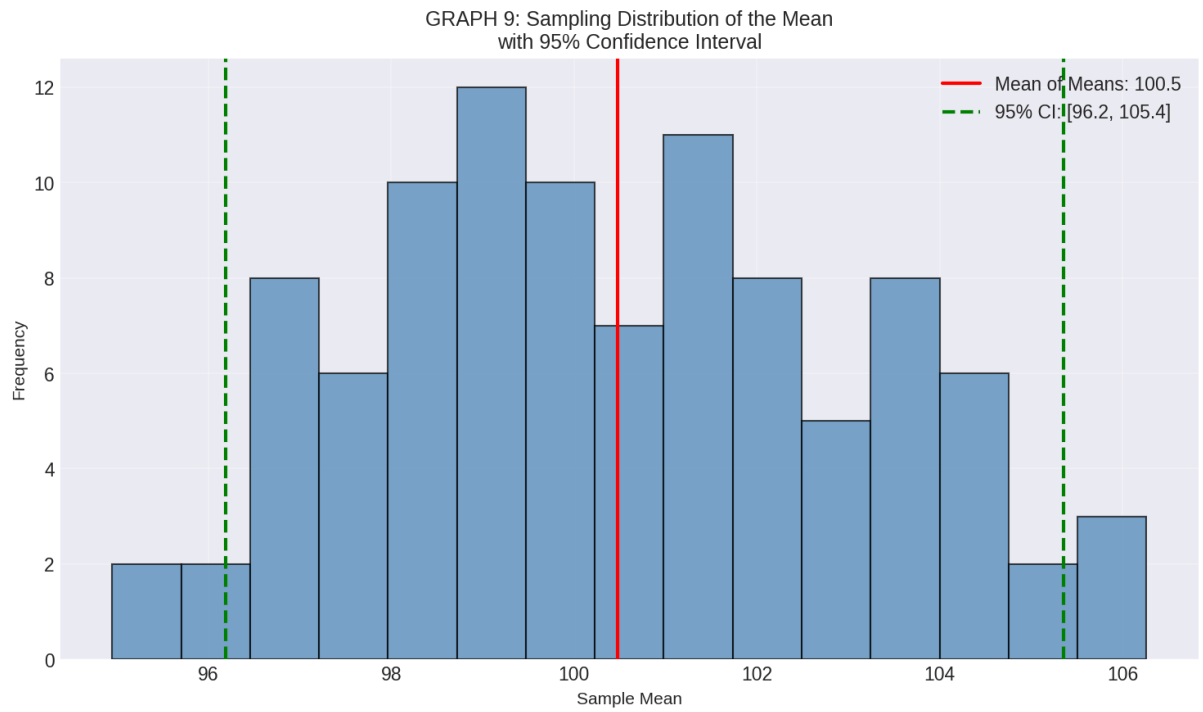

Figure 21: Sampling Distribution with 95% Confidence Interval